

**GANDHI INSTITUTE OF ENGINEERING
AND TECHNOLOGY UNIVERSITY,
ODISHA, GUNUPUR**

DEPARTMENT

OF

**COMPUTER SCIENCE &
ENGINEERING**

(SCHOOL OF ENGINEERING & TECHNOLOGY)



BACHELOR'S DEGREE PROGRAMME

B. Tech – CSE(AIML)

Course Structure and Syllabus

Academic Session

[2023-27]



GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1. Institute Vision & Mission, Department Vision & Mission, PEO's, PO's, PSO's

Institute Vision:

To be a renowned and Globally Recognized University giving importance to academic excellence with latest technology, Research, Innovation and Entrepreneurial Attitude.

Institute Mission:

- To create innovative and committed work force to cater to the societal, environmental and economic needs of the Nation.
- To promote Education and Research Globally at par with the international standards
- To prepare the future leaders with latest skills befitting to become Entrepreneurs or Employable.
- To support and uplift the meritorious students of this Tribal Area to represent as an Ambassador in all forums as a part of our social responsibility.

Department Vision:

To be globally acclaimed as a premier Computer Science and Engineering department through excellent teaching and strong research environment by creating professionally competent engineers with moral values committed to build a vibrant nation.

Department Mission:

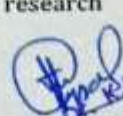
- Provide a positive and professional learning environment to master the fundamental principle of computing, problem solving skills, leadership qualities, team-spirit, ethics and societal responsibilities.
- To enable the graduates to use modern tools and technologies in the area of computer science and information technology.
- Establish Industry Institute Interaction program to enhance the employability skills.
- To promote innovative research in Computer Science and Engineering to serve the needs of Industry, Government and Society at large.

Program Educational Objectives:

- To render best platform for adequate training and opportunities to work as teams on projects with effective communication skills and leadership qualities and understand professional ethics, social awareness and organizational context in which their engineering skills are utilized.

To endow the students with sound knowledge in the field of mathematics, basic science and engineering fundamentals to solve and inculcate the ability to utilize their skills to prepare them for higher studies, research and analyze engineering problems.




R. MURALI GOPAL
MOD. CSE
GIET UNIVERSITY
GUNUPUR-765022

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- To extend an ability to analyze the need of the society by providing innovation solutions, leading to their personal cum professional growth as an entrepreneur.

Program Outcome:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

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- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes:

- To provide students an understanding of the expectations of industry and practical competence with a broad range of programming language and open source platforms through value added courses.
- The ability to analyze and develop computer programs in the areas related to Artificial Intelligence, Big Data Analysis and Cyber Security for efficient design of computer-based systems of varying complexity.

Program Educational Objectives:

- Develop proficiency in the field of mathematics, basic science, and engineering fundamentals to analyze and solve a wide range of computational problems in industry or other work environments and succeed in an advanced degree program in a field such as engineering.
- Possess the ability to collaborate as team members and team leaders for facilitating technical solutions for computing systems with awareness and commitment to their ethical responsibilities.
- Attain the ability to critically analyze the need of society and provide innovative solutions in multi-disciplinary areas adopting technologies leading to their personal and professional growth as an entrepreneur.

III Semester [Second Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	BS	23BBSBS23003	Applied Statistics	3	1	0	4	40	60
2	PC	23BCMPC23001	Data Base Management Systems	3	0	0	3	40	60
3	ES	23BECES23001	Digital Electronics	3	0	0	3	40	60
4	PC	23BCMPC23002	Object Oriented Programming through Java	3	0	0	3	40	60
5	PE	23BCMPE23011 /23BCMPE23011	PE-I (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS23001	Intermediate communication skills and critical thinking	1	0	0	1		
			TOTAL	16	1	0	17	200	300
PRACTICAL/SESSIONAL									
7	ES	22BECES23101	Digital Electronics Circuits Lab	0	0	2	1	30	20
8	PC	22BCMPC23102	Object Oriented Programming through Java Lab	0	0	2	1	30	20
9	PC	22BCMPC23101	Data Base Management Systems Lab	0	0	2	1	30	20
10	HS	23BBSHS23101	Intermediate communication skills and critical thinking	0	0	2	1	30	20
11	PST	23BCSPW23101	Summer Internship-I	0	0	0	1	30	20
12	PST	23BCSLP23101	Learning Project-I	0	0	0	1	50	0
			TOTAL	0	0	8	6	200	100
			TOTAL	16	1	8	23	400	400

Professional Elective Bucket

Sl.	Sem	Professional Elective I
1	23BCMPE23011	Artificial Intelligence & Expert Systems
2	23BCMPE23012	Linear and Non-Linear Programming

Subject Code	Name of the Subject	L	T	P	C									
	Object Oriented Programming Through Java	3	0	2	4									
Course Educational Objectives														
CEO 01	The model of object oriented programming: abstract data types, encapsulation, inheritance and polymorphism													
CEO 02	Fundamental features of an object oriented language like Java: object classes and interfaces, exceptions and libraries of object collections													
CEO 03	How to take the statement of a business problem and from this determine suitable logic for solving the problem; then be able to proceed to code that logic as a program written in Java.													
CEO 04														
Course Outcomes: Upon successful completion of this course, students should be able to:														
CO 01	Knowledge of the structure and model of the Java programming language.													
CO 02	Use the Java programming language for various programming technologies.													
CO 03	Develop software in the Java programming language.													
CO 04	Evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements.													
CO 05	Propose the use of certain technologies by implementing them in the Java programming language to solve the given problem.													
CO 06	Choose an engineering approach to solving problems, starting from the acquired knowledge of programming.													
CO-PO & PSO Mapping (Articulation Matrix):														
COs	PROGRAMME OUTCOMES												PSOs	
	01	02	03	04	05	06	07	08	09	10	11	12	01	02
01	3	1											1	
02	3	2											1	
03	2	2	2										2	
04	2	1	2										1	
05	2	1	3										1	
06		3	3										1	
Avg.	2	1.6	2.5										1.16	
Syllabus														
UNIT I													13 Hours	
An introduction to Object Oriented Programming, Features of Object Oriented Programming Introduction to Java. Difference between C/C++ and Java, Features of Java, First Java Program, Writing the java program, Compiling the program, JVM and its significance in executing a program, Architecture of JVM. Taking Input from keyboard, Command Line Arguments, Using Scanner Class, Using Buffered Reader class. Java Tokens, Datatypes, Operators, Control Structures, Conditional Statements, Iteration Statement/Loops, Jumping Statements, Java Arrays, 1-dimensional arrays, 2-dimensioanl arrays.														
UNIT II													15 Hours	
Introduction to Classes and Objects. Constructors, static Keyword, this Keyword, Array of Objects, Access Modifiers (Public, Private, Protected, Default). Inheritance, Types of Inheritance and Java-supported Inheritance, super, Polymorphism, Method Overloading, Constructor Overloading, Method Overriding, Dynamic Method Dispatching. Interfaces, Multiple Inheritance Using Interfaces, abstract classes. String Handling in Java: String, StringBuffer, StringBuilder, StringTokenizer Wrapper classes, Auto boxing and unboxing.														

UNIT III		12 Hours
Java API Packages, User-Defined Packages, Accessing Packages. Error and Exception Handling, Types of exceptions Hierarchy of Exception classes, try, catch, finally, throw, throws, Commonly used Exceptions and their details ,User defined exception classes. Multithreading, Thread in Java, Thread execution prevention methods. (yield(), join(), sleep()), Concept of Synchronization, Inter Thread Communication, Basics of Deadlock, Demon Thread. Inner Classes: Introduction, Member inner class, Static inner class, Local inner class, Anonymous inner class. IO Streams (java.io package), Byte Stream and Character Stream, Files and Random Access Files, Serialization.		
UNIT IV		12 Hours
Collection Frame Work (java.util), Util Package interfaces, List, Set, Map. Applet Introduction, Life Cycle of an Applet, GUI with an Applet. AWT : The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, Scroll Bar. Working with Frame class, Colour, Fonts and layout managers. Event Handling: Events, Event sources, Event Listeners, Event Delegation Model (EDM), Handling Mouse and Keyboard Events, Adapter classes		

Subject Code	Name of the Subject	L	T	P	C										
	Database Management System	3	0	1	4										
Course Educational Objectives															
Pre -Requisite: A student should have basic idea on programming and file systems.															
CEO 1	Gain a good understanding of the architecture and functioning of Database Management Systems as well as associated tools and techniques.														
CEO 2	Understand and apply the principles of data modeling using Entity Relationship and develop a good database design.														
CEO3	Understand the use of Structured Query Language (SQL) and its syntax														
CEO4	Apply Normalization techniques for effective database design.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify and Classify the concepts of Database Management system, Data models and architecture of database, ER to Relational mapping concepts.														
CO2	Applying The constraints in database using different query languages like:- relational algebra and calculus implementing the Data definition and data manipulate languages in Database.														
CO3	Compare the different normal forms to Apply normalization process to construct consistent Database.														
CO4	Design and Develop the Database with concurrency control and recovery strategies to make complete Database without confliction and anomalies in concurrent access environment.														
CO5	Develop efficient storage scheme of saving and retrieving Records and Files														
CO6	Apply the principles of database transaction management, database recovery and security.														
CO-PO & PSO Mapping:															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1	1												
CO2	3	2	1												
CO3	2	1	1												
CO4	2	1	2												
CO5	2	1	2												
CO6	3	2	1												
Avg.	2.5	1.33	1.33												
SYLLABUS															

UNIT:1	(6 Hours)
Introduction to database Systems: advantages of database system over traditional file system, Basic concepts & Definitions, Database users, Database Language, Database System Architecture, Schemas, Sub Schemas, & Instances, database constraints, 3-level database architecture, Data Abstraction, Data Independence, Mappings, Structure, Components & functions of DBMS, Data models.	
Entity-Relationship model: Basic concepts, Design process, Constraints, Keys, Design issues, E-R Diagrams, weak entity sets, Extended E-R features – generalization, specialization, aggregation, Reduction to E-R database schema.	
UNIT:2	(10 Hours)
Relational Query Languages: Relational Operations. Relational Algebra, Selection and projection set operations, renaming, Joins, Division, Examples of Algebra overviews, Relational calculus, Tuple relational Calculus, Domain relational calculus, Extended relation algebra operation	
UNIT:3	(10 Hours)
Normalization: Introduction, non-loss decomposition and functional dependencies, First, Second, and third normal forms – dependency preservation, Boyce-Codd normal form. Higher Normal Forms - Introduction, Multi-valued dependencies and fourth normal form, Join dependencies and fifth normal form	
UNIT:4	(10 Hours)
Transaction Concept: Transaction properties (ACID), Transaction State, Schedules, Serializability (Conflict Serializability and View Serializability), Need for Concurrency, Concurrency Control- Lock based control, time stamping Protocol, Two Phase Locking, Dead lock in DBMS	
Recovery and Atomicity: Log Based Recovery, Transaction Logs and Shadow Paging, Types of Recovery Techniques in DBMS, Rollback/Undo Recovery Technique, Commit/Redo Recovery Technique, Checkpoint Recovery, Backup Techniques	
UNIT:5	(6 Hours)
Detailed Storage Architecture: Storage Strategies: Detailed Storage Architecture, Storing Data, Magnetic Disk, RAID, Other Disks, Magnetic Tape, Storage Access, File & Record Organization, File Organizations & Indexes, Order Indices, B+ Tree Index Files, Hashing Data Dictionary	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Sudarshan, Korth: Database System Concepts, 6th edition, McGraw-Hill Education 2. Elmasari&Navathe: Fundamentals of Database System, Pearson Education.	
Reference Books:	
1. Elmasari&Navathe: Fundamentals of Database System, Pearson Education. 2. Ramakrishnan: Database Management Systems, McGraw-Hill Education. 3. Andrew S. Tanenbaum: Modern Operating Systems, 3rd Edition, Pearson Education. 4. Terry Dawson, Olaf Kirch: Linux Network Administrator's Guide, 3rd Edition O'Reilly	

Subject Code		SUBJECT NAME											L	T	P	C
		Artificial Intelligence and Expert Systems											3	0	0	3
Course Educational Objectives																
CEO 01	To understand Basic Concepts of Artificial Intelligence and Expert Systems.															
CEO 02	To provide knowledge on Various Techniques and Tools involved in Artificial Intelligence.															
CEO 03	Explain about AI techniques for knowledge representation, planning and uncertainty Management.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO 01	Understand the Basics about Artificial Intelligence and Expert Systems.															
CO 02	Understand the Programming Logics in Artificial Intelligence.															
CO 03	Develop knowledge of decision making and learning methods															
CO 04	Demonstrate the latest developments in Knowledge systems and Tools.															
CO 05	Describe the use of AI to solve NLP problems.															
CO 06	Understand the Knowledge about the Expert Systems.															
CO-PO & PSO Mapping (Articulation Matrix):																
COs	PROGRAMME OUTCOMES												PSOs			
	01	02	03	04	05	06	07	08	09	10	11	12	01	02		
01	3	1	2													
02	3	2	3													
03	2	3	2													
04	3	3	2	3												
05	3	3	3													
06	3	2	2													
Avg.	2.83	2.33	2.33	3												
Justification of the Mapping:																
CO	PO	Level	Justification													
1	1	3	It has the highest significance													
	2	1	It has the least significance													
	3	2	It has an average significance													
2	1	3	It has the highest significance													
	2	2	It has an average significance													
	3	3	It has the highest significance													
3	1	2	It has an average significance													
	2	3	It has the highest significance													
	3	2	It has an average significance													
4	1	3	It has the highest significance													
	2	3	It has the highest significance													
	3	2	It has an average significance													
	4	3	It has the highest significance													
5	1	3	It has the highest significance													
	2	3	It has the highest significance													
	3	3	It has the highest significance													
6	1	3	It has the highest significance													
	2	2	It has an average significance													
	3	2	It has an average significance													
Syllabus																

UNIT I		10 Hours
Introduction to AI: What is Artificial Intelligence? AI Technique, Types of AI (Strong AI & Weak AI), Agents and Environments, Level of the Model, Problem Spaces, and Search: Defining the Problem as a State Space Search-Water Jug Problem, Production Systems, Problem Characteristics, Production System Characteristics, and Issues in the Design of Search Programs.		
UNIT II		08 Hours
Knowledge Representation: Representations and Mappings, Approaches to Knowledge Representation. Using Predicate Logic: Representing Simple Facts in Logic, Representing Instance and ISA Relationships, Computable Functions and Predicates, Resolution, Natural Deduction. Using Rules: Procedural Versus Declarative Knowledge, Logic Programming, Forward Versus Backward Reasoning, Matching, Control Knowledge. Symbolic Reasoning Under Uncertainty: Introduction to Non monotonic Reasoning, Logics for Non monotonic Reasoning, Implementation Issues, Augmenting a Problem-solver, Depth-first Search, and Breadth-first Search.		
UNIT III		08 Hours
Game Playing: The Mini-max Search Procedure, Adding Alpha-beta Cutoffs, Iterative Deepening. Planning: The Blocks World, Components of a Planning System, Goal Stack Planning, Nonlinear Planning Using Constraint Posting, Hierarchical Planning, Other Planning Techniques. Understanding: What is Understanding, What Makes Understanding Hard?, Understanding as Constraint Satisfaction.		
UNIT IV		08 Hours
Natural Language Processing: Introduction, Syntactic Processing, Semantic Analysis, Discourse and Pragmatic Processing, Statistical Natural Language Processing, Spell Checking. Learning: Rote Learning, Learning by Taking Advice, Learning in Problem-solving, Learning from Examples: Induction, Explanation-based Learning, Discovery, Analogy, Formal Learning Theory, Neural Net Learning and Genetic Learning.		
UNIT V		08 Hours
Expert Systems: Architecture and Roles of Expert System-Typical Expert System Representing and Using Domain Knowledge, Expert System Shells, Explanation, Knowledge Acquisition.		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books: 1. Elaine Rich, Kevin Knight & Shivashankar B Nair, <i>Artificial Intelligence</i>, McGraw Hill, 3rd ed., 2009 2. Stuart Russell, Peter Norvig, "Artificial Intelligence – A Modern Approach", 3rd Edition, Pearson Education / Prentice Hall of India, 2010. 3. Joseph C. Giarratano, Gary D. Riley, "Expert Systems: Principles and Programming", 4th Edition, 2015		
Reference Book: 1. S K Aushik, <i>Artificial Intelligence</i>, Cengage Learning, 1st ed., 2011 2. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI., 2011 3. Nils J. Nilsson, "Artificial Intelligence: A new Synthesis", Harcourt Asia Pvt. Ltd., 2000. 4. Janakiraman, K. Sarukesi, 'Foundations of Artificial Intelligence and Expert Systems', Macmillan Series in Computer Science, 2000.		

Subject Code	Name of the Subject										L	T	P	C	
	Applied Statistics										3	1	0	3	
Course Educational Objectives															
Pre-Requisite: A student should have basic idea on programming and file systems.															
CEO 1															
CEO 2															
CEO 3															
CEO 4															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify the role that statistics can play in the engineering problem-solving process, discuss the different methods that engineers use to collect data and, construct and interpret visual data displays.														
CO2	Compute and interpret the descriptive statistics, correlation coefficient and rank correlation coefficient, use simple linear regression model to engineering data.														
CO3	Explain various sampling methods, compute and explain point estimators and interval estimators for mean, variance and proportion														
CO4	Structure engineering decision-making problems as hypothesis tests, use z-test, t-test, chi-square and Ftest to test the statistical hypotheses, p-value approach for making decisions in hypothesis tests, explain and use the relationship between confidence interval and hypothesis tests.														
CO-PO & PSO Mapping:															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	2	2	2												
CO2	2	3	2												
CO3	2	1	2												
CO4	2	2	2												
Avg.	2	2	2												
SYLLABUS															
UNIT:1 (10 Hours)															
The Role of Statistics in Engineering : The Engineering Method and Statistical Thinking - Collecting Engineering Data - Basic Principles - Retrospective Study - Observational Study - Designed Experiments -Observing Processes Over Time - Mechanistic and Empirical Models															
Data Description and Representation: Collection of data- Classification and Tabulation of data - Stem-and-Leaf Diagrams - Frequency Distributions and Histograms - Box Plots - Time Sequence Plots - Probability Plots															
UNIT:2 (10 Hours)															
Descriptive Statistics: Measures of central Tendency-Measures of Dispersion- Skewness and Kurtosis.															
Correlation and Regression: Scatter Diagram – Types of Correlation – Karl Pearsons Coefficient of Correlation and Spearmen’s Rank Correlations- Method of Least Squares – Linear Regression.															

UNIT:3 (12 Hours) Sampling: Different types of sampling - Sampling Distributions – Sampling Distribution of Mean. Point Estimation of Parameters: General Concepts of Point Estimation - Unbiased Estimators - Variance of a Point Estimator - Standard Error- Methods of Point Estimation (Method of Moments - Method of Maximum Likelihood). Statistical Intervals for a Single Sample: Confidence Interval on the Mean of a Normal Distribution with Variance Known - Confidence Interval on the Mean of a Normal Distribution with Variance Unknown - Confidence Interval on the Variance and Standard Deviation of a Normal Distribution - A Large-Sample Confidence Interval for a Population Proportion.
UNIT:4 (12 Hours) Tests of Hypotheses for a Single Sample: Tests of Statistical Hypotheses - General Procedure for Hypothesis Testing –Tests on the Mean of a Normal Distribution with Variance Known - Tests on the Mean of a Normal Distribution with Variance Unknown - Tests on the Variance and Standard Deviation of a Normal Distribution . Statistical Inference for Two Samples: Inference For a Difference in Means of Two Normal Distributions with Variances Known - Inference For a Difference in Means of Two Normal Distributions with Variances Unknown -Inference on the Variances of Two Normal Distributions – Inference on Two Population Proportions.
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. Douglas C. Montgomery and George C. Runger. <i>Applied Statistics and Probability for Engineers</i>, (3rdEdn), John Wiley and Sons, Inc., New York, 2003. 2. Robert H. Carver and Jane Gradwohl Nash. <i>Doing Data Analysis with SPSS Version 18.0</i>, (Indian Edition), Cengage Learning, New Delhi, 2012. 3. Richard A. Johnson and C.B.Gupta, <i>Probability and Statistics for Engineers</i>, (7thEdn.), Pearson Education, Indian Impression 2006.
Reference Books: 1. Mohammed A.Shayib. <i>Applied Statistics, First Edition. EBook, Bookboon.com</i> 2013. 2. Peter R.Nelson, Marie Coffin, Copeland Kanen, A.F. <i>Introductory Statistics for Engineering Experimentation, Elsevier Science and Technology Books</i>, New York, 2003. 3. Sheldon M. Ross, <i>Introduction to Probability and Statistics</i>, (3rdEdn), Elsevier Science and Technology Books, New York, 2004. 4. T.T.Soong, <i>Fundamentals of Probability and Statistics for Engineers</i>, John Wiley and Sons, Ltd., New York, 2004. 5. J.P.Marques de Sá , <i>Applied Statistics using SPSS, STATISTICA, MATLAB and R</i>, (2ndEdn.), Springer Verlag, Heidelberg, 2007.
Online resources ➤ http://apus.libguides.com ➤ www.mii.lt ➤ www.sosmath.com ➤ www.macalester.edu ➤ www.nptel.iitm.ac.in ➤ www.purplemath.com

		DIGITAL ELECTRONICS	3-0-0
Pre-requisites (if any):		Knowledge of Basic Mathematics & Basic Electrical and Electronics Engineering.	
Course Objectives:			
–	To understand digital logic fundamentals, including binary systems and Boolean algebra.		
–	To design and analyze combinational and sequential circuits, including logic gates, flip-flops, and counters.		
–	To apply digital electronics in real-world applications, using simulation tools and programmable logic devices (PLDs).		
Course Content			
Unit 1			(10 Hours)
Number Systems and Codes: Binary, Octal, Hexadecimal, and Decimal Number System and their Conversion; Representation of Signed Binary and Floating-Point Number; Binary Arithmetic using 1’s and 2’s Complements, Binary Codes - BCD Code, Gray Code, ASCII Character Code.			
Boolean Algebra and Logic Gates: Axioms and Laws of Boolean Algebra; Reducing Boolean Expressions; Logic Levels and Pulse Waveforms; Logic Gates; Boolean Expressions and Logic Diagrams; Canonical and Standard Forms.			
Unit 2			(12 Hours)
Gate-level Minimization: K-maps - Two, Three, and Four Variable K-maps, Don’t-Care Conditions; NAND and NOR Implementation; Other Two-Level Implementations, Exclusive-OR Function.			
Combinational Logic: Combinational Circuits; Analysis Procedure; Design Procedure; Adders; Subtractors; Parallel Binary Adders; Binary Adder-Subtractor; Binary Multiplier; Magnitude Comparator; Decoders; Encoders; Multiplexers; De-multiplexers.			
Unit 3			(8 Hours)
Synchronous Sequential Logic: Sequential Circuits; Storage Elements: Latches, Flip-Flops, Master-Slave Flip-Flop; Conversion of Flip-Flops; Analysis of Clocked Sequential Circuits; Mealy and Moore Models of Finite State Machines; Design Procedure.			
Unit 4			(8 Hours)
Registers and Counters: Shift Registers; Data Transmission in Shift Registers; SISO, SIPO, PISO, and PIPO Shift Registers; Counters; Asynchronous Counters; Design of Asynchronous Counters; Synchronous Counters; Design of Synchronous Counters; Ring Counter.			
Unit 5			(6 Hours)
Memory and Programmable Logic: Introduction; Random-Access Memory; Memory Decoding; Error Detection and Correction; Read-Only Memory; Programmable Logic Array; Programmable Array Logic; Sequential Programmable Devices.			
Text Books:			
[1] M. Morris Mano, and Michael D. Ciletti, Digital Design: With an Introduction to			

- the Verilog HDL, VHDL, and SystemVerilog*. Pearson Education, Sixth Edition, 2017.
- [2] A. Anand Kumar, *Fundamentals of Digital Circuits*. PHI Learning Pvt. Ltd., New Delhi, Fourth Edition, 2016.
- [3] R. P. Jain, *Modern Digital Electronics*. Tata McGraw-Hill Education Pvt. Ltd., Fourth Edition, 2009.

Reference Books:

- [1] John P. Uyemura, *A First Course in Digital Systems Design: An Integrated Approach*. Thomson Press (India) Ltd., India Edition, 2002.
- [2] William H. Gothmann, *Digital Electronics: An Introduction to Theory and Practice*, PHI Learning Pvt. Ltd., New Delhi, Second Edition, 2006.
- [3] D.V. Hall, *Digital Circuits and Systems*. Tata McGraw-Hill Education Pvt. Ltd., 1989.
- [4] Charles H. Roth, *Digital System Design using VHDL*. Tata McGraw-Hill Education Pvt. Ltd., Second Edition, 2012.

Course Outcomes:

At the end of this course, students will be able to

- CO1: Define and analyze the number systems and conversions.
- CO2: Perform binary arithmetic with 1's and 2's complements and understand binary codes.
- CO3: Apply Boolean algebra laws to simplify expressions and design logic circuits.
- CO4: Use K-maps for logic minimization and design combinational circuits.
- CO5: Analyze and design synchronous sequential circuits.
- CO6: Design shift registers, and counters, and understand RAM, ROM, and PLDs.

CO-PO & PSO matrices of Course

[illegible]

DIGITAL ELECTRONICS LABORATORY		0-0-2
Pre-requisites (if any):	Knowledge of Basic Electrical and Electronics Engineering.	
Course Objectives:		
–	To gain hands-on experience with basic logic gates and combinational circuits.	
–	To implement and test combinational and sequential circuits.	
–	To utilize simulation tools for circuit design and testing.	
Course Content		
List of Experiments:		
(At least 10 experiments should be done, Experiments No. 1 and 2 are compulsory, and out of the balance 8 experiments at least 3 experiments have to be implemented through both Verilog/VHDL and hardware implementation as per the choice of the student totaling to 6 and the rest 2 can be either through Verilog/VHDL or hardware implementation.)		
1. Digital Logic Gates: Investigate the logic behavior of AND, OR, NAND, NOR, EX-OR, EXNOR, Inverter, and Buffer Gates, use of Universal NAND/NOR Gate. 2. Gate-Level Minimization: Two-level and multi-level implementation of Boolean functions. 3. Combinational Circuits: Design, assemble, and test: adders and subtractors, code converters, gray code to binary, and 7-segment displays. 4. Design, implement, and test a given design example with (i) NAND Gates only, (ii) NOR Gates only, and (iii) using the minimum number of Gates. 5. Design with Multiplexers and De-multiplexers. 6. Flip-Flop: Assemble, test, and investigate the operation of SR, D & JK flip-flops. 7. Shift Registers: Design and investigate the operation of all types of shift registers with parallel load. 8. Counters: Design, assemble, and test various ripple and synchronous counters - decimal counter, Binary counter with the parallel load. 9. Memory Unit: Investigate the behavior of the RAM unit and its storage capacity - 16×4 RAM: testing, simulating, and memory expansion. 10. Clock-Pulse Generator: Design, implement, and test. 11. Parallel Adder and Accumulator: Design, implement, and test. 12. Binary Multiplier: Design and implement a circuit that multiplies 4-bit unsigned numbers to produce an 8-bit product. 13. Verilog/VHDL simulation and implementation of Experiments listed in Sl. No. 3 to 12.		

Course Outcomes:

At the end of this course, students will be able to

CO1: Investigate the behavior and application of various digital logic gates.

CO2: Implement two-level and multi-level Boolean function minimization using universal gates like NAND and NOR.

CO3: Design, assemble, and test combinational circuits.

CO4: Assemble, test, and investigate flip-flops and shift registers, and design various types of counters

CO5: Investigate RAM behavior and design memory units.

CO6: Use Verilog/VHDL for simulation and implementation of combinational and sequential circuit designs.

CO-PO & PSO matrices of Course

[illegible]

Subject Code		Name of the Subject										L	T	P	C	
		Intermediate Communication Skills and Critical Thinking										3	0	0	3	
Course Educational Objectives																
CEO1	To develop the communication skills and soft skills of the students.															
CEO2	To enhance employability and entrepreneurial skills.															
CEO3	To promote critical thinking.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	Recognise the importance of LSRW skills.															
CO2	Develop effective listening skills.															
CO3	Demonstrate public speaking skills.															
CO4	Distinguish professional writing from general writing.															
CO5	Develop effective reading skills.															
CO6	Execute critical thinking.															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	2	1	2													
CO2	2	1	3													
CO3	2	2	3													
CO4	2	2	3													
CO5	2	2	1													
CO6			2													
Avg.	2	2	2.3													
SYLLABUS																
UNIT-1 Comprehensive Listening Skills (4 Hours)																
1.1 What and why																
1.2 Techniques for improving listening skills																
UNIT-2 Public Speaking Skills (4 Hours)																
2.1 Public speaking; what and why																
2.2 Overcoming stage fear																
2.3Techniques for effective speaking skill																
UNIT- 3 Stages of Writing (4 Hours)																
3.1 Pre-Writing- selecting a topic																
3.2 Writing- Free Writing, Drafting																
3.3 Re-writing- Revision, editing, proof Reading																
UNIT- 4 Effective Reading Skills (3 Hours)																
4.1 What is Reading Comprehension																
4.2 Tips for effective reading																
4.3Techniques and types.																
UNIT-5 Critical Thinking Skills (4 Hours)																
5.1 What and Why																
5.2 Critical thinking vs. Lateral thinking																
5.3 Ways to improve Critical Thinking																

Teaching Methods: Chalk & Board/PPT/Video Lectures
Reference Books: 1. Basic Communication Skills for Technology by Andrea J. Rutherford, Pearson Education Asia, Patparganj, New Delhi, 2001. 2. How to Read better and Faster by Norman Lewis, 4th Edition, Publisher: Crowell, 1978. 3. The Kabuliwala and other stories by Rabindranath Tagore, Maple Classics, 2019. 4. The Gift of Magi by O Henry, Candlewick Press, 2008.

Activity- Answering hypothetical question in a mock interview	
Video Resume	(2 hours)
Activity- Preparing a digital resume	
Professional e-mail	(2 hours)
Activity- Writing e-mail for professional purpose	

Minutes of the Meeting	(2 hours)
Activity- Preparing minutes of mock meetings.	
Memo Writing	(2 hours)
Writing Memos for internal organization	
Vocabulary Exercise	(2 hours)
Activity- Playing word games in small groups.	
Intermediate Reading Comprehension	(2 hours)
Activity-Reading an Intermediate level passage/ poems and answer questions.	
Preparing a Sales Pitch	(2 hours)
Activity- Prepare a presentation on any product (real/imaginary) of choice and present a sales pitch	
Critical Thinking Activity	(2 hours)
Activit- critically analysis of Kabuliwala by Rabindranath Tagore OR The Gift of Magi by O Henry	
Critical Thinking	(2 hours)
Activity 2- Critically Analyse Stopping by the Woods in a Snowy Evening by Robert Frost OR Ozymandias by P.B Shelly.	
Reference Books 1. An Introduction to Professional English and Soft Skills by B. K. Das et al., Cambridge University Press, 2009. 2. Communicative English for Engineers and Professionals by Nitin Bhatnagar and Mamta Bhatnagar. Published by DK/Pearson, 2010. 3. Practical English Usage . Michael Swan, OUP, 1995	

IV Semester [Second Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

S N	CC	Sub Code	Name of the Subject	L	T	P	C	IS A	ES A
THEORY									
1	PC	23BCMPC24 001	Python Programming for Machine Learning	3	0	0	3	40	60
2	PC	23BCMPC24 002	Computer Organization & Architecture	3	0	0	3	40	60
3	PC	23BCMPC24 003	Design and Analysis of Algorithms	3	0	0	3	40	60
4	PC	23BCMPC24 004	Operating Systems	3	0	0	3	40	60
5	PE	23BCMPE24 011 /23BCMPE2 4012	PE-II (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS24 001	English Learning -II (Advance communication skills and professional Ethics)	1	0	0	1		
			TOTAL	1 6	0	0	1 6	20 0	30 0
PRACTICAL/SESSIONAL									
7	PC	22BCMPC24 101	Python Programming Lab	0	0	2	1	30	20
8	PC	22BCMPC24 103	Design and Analysis of Algorithms Lab	0	0	2	1	30	20
9	PC		Modeling and Analytics Lab	0	0	2	1	30	20
1 0	HS	23BBSHS24 101	English Learning -II (Advance communication skills and professional Ethics) Lab	0	0	0	1	50	0
1 1	PST	23BCSLP24 101	Learning Project-II	0	0	0	1	50	0
			TOTAL	0	0	6	5	19 0	60
			TOTAL	1 6	0	6	2 1	39 0	36 0

Professional Elective Bucket

Sl .	Sem	Professional Elective II
1	23BCMPE2 4011	Introduction to Machine Learning
2	23BCMPE2 4012	Fuzzy Logic & Applications
3		

Subject Code			Name of the Subject									L	T	P	C	
			Python Programming for Machine Learning									3	1	0	4	
Course Educational Objectives																
CEO1	To understand the basics of programming using Python.															
CEO2	To Construct and execute basic programs in Python.															
CEO3	To implement database connectivity, CGI, Multithreading and web development in python.															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Understand the basic concept of Python like data types and syntax of programming.															
CO2	use python programming with user defined & built in methods, objects of Python.															
CO3	design application using the concepts of file, database access and exception handling.															
CO4	create practical and contemporary applications such as web applications and discrete-event simulations, data analysis and IoT devices.															
CO 5	apply Python programming techniques to implement various machine learning algorithms, including supervised and unsupervised learning models, to solve real-world problems.															
CO 6	evaluate the performance of machine learning models by analyzing key metrics and fine-tuning hyperparameters to optimize the models' accuracy and efficiency.															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES															
	1	2	3	4	5	6	7	8	9	10	11	12				
CO1	3	1										1				
CO2	2	3	1	3	2							1				
CO3	2	3	3	3	2				1		1	2				
CO4	2	3	3	3	2				1		1	2				
CO5	2	2	3	3	2				1							
CO6	2	3	3	3	2				1							
Avg.	2.25	2.5	2.6	3	2				1		1	1.5				
SYLLABUS																

UNIT:1	(08 Hours)
Introduction: Installation, First Python Program: Interactive Mode Programming, Script Mode Programming; Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation & Comments; Assigning Values to Variables, Multiple Assignment. Standard Data Types: Numbers, Strings, Lists, Tuples, Dictionary; Data Type Conversion; Basic Operators: Arithmetic, Comparison, Assignment, Bitwise; Operators: Logical, Membership, Identity; Operators Precedence; Python Numbers & Mathematical functions, Python Strings.	
UNIT:2	(10 Hours)
Python Program Flow Control: Conditional blocks using if, else and elif, Simple for loops in python, For loop using ranges, string, list and dictionaries, Use of while loops in python, Loop manipulation using pass, continue, break and else, Programming using Python conditional and loops block Python Functions, Modules and Packages: Organizing python codes using functions,organizing python projects into modules, Importing own module as well as external modules, Understanding Packages, Powerful Lamda function in python, Programming using functions, modules and external packages Python String, List and Dictionary Manipulations: Building blocks of python programs, understanding string in build methods, List manipulation using in build methods, Dictionary manipulation, Programming using string, list and dictionary in build functions	
UNIT:3	(12 Hours)
Python OOP: Object-Oriented Programming (OOP) concepts, Classes and objects in Python Inheritance and polymorphism. Python Libraries for Data Science: Introduction to data science libraries in Python: NumPy, Pandas, and Matplotlib, Data types and data structures in NumPy, Data manipulation with Pandas, Data visualization with Matplotlib. Data Visualization: Introduction to data visualization, Matplotlib library for visualization, Seaborn library for visualization.	
UNIT:4	(15 Hours)
Introduction to Machine Learning: Overview of Machine Learning, Supervised learning, Unsupervised learning, Regression, Clustering. Machine Learning with Scikit-Learn: Introduction to Scikit-Learn, Regression and classification, Clustering, Model evaluation and validation.	
UNIT:5	(15 Hours)
Deep Learning with Python: Introduction to Deep Learning, Neural Networks in Python, Deep Learning frameworks: Keras, TensorFlow, and PyTorch, Convolutional Neural Networks (CNNs) in Python. Natural Language Processing with Python: Introduction to Natural Language Processing (NLP), Text Preprocessing in Python, NLP Libraries in Python: NLTK, Spacy, and TextBlob, Sentiment Analysis in Python.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. “Python: The Complete Reference”, Martin C. Brown, McGraw-Hill/Osborne Media. 2. “Learning Python”, Mark Lutz, O’Reilly Media, Inc., Fifth Edition, 2013.	

Reference Books:

1. “Expert Python Programming”, Tarek Ziad, Packt Publishing.
2. “Python Essential Reference, David Beazley”, Developer's Library, Sams Publishing.

Subject Code		Name of the Subject										L	T	P	C	
		Computer Organization and Architecture										3	0	0	3	
Course Educational Objectives																
CEO1	Identify the functional units in a digital computer system,															
CEO2	Distinguish between the various ISA styles and trace the execution sequence of an instruction through the processor.															
CEO3	Compare different approaches used for implementing a functional unit and evaluate different computer systems based on performance metrics.															
CEO4	Provide an outline of working principles of components of computer.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Explain and illustrate the execution cycle of a program.															
CO2	Classify the components of digital computer and designing of control and memory units with performance evaluation.															
CO3	Compute different types of binary arithmetic operation. .															
CO4	Identify the characteristics of memories and explain memory and IO devices working process.															
CO5	Classify the concept of Parallelism															
CO6	Explain memory and types and interrupts															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	1	1			2											
CO2	2	1			3											
CO3	3	2			2											
CO4	1	1			1											
CO5	2	2			2											
CO6	1	3			2											
Avg.	1.6	1.6			2											
SYLLABUS																
UNIT:1 (12 Hours)																
Functional Units of a computer, Operational concepts, Bus structures, Memory locations and addresses, Memory operations, Instructions, Instruction sequencing, addressing modes, Problem solving, Introduction to Microprocessor, Introduction to Assembly language, writing of assembly language programming, ARM Processor: The thumb instruction set, Processor and CPU cores, Instruction Encoding format, Memory load and Store instruction in ARM, Basics of IO operations.																
UNIT: 2 (12 Hours)																
Addition and subtraction of Signed numbers, Problem solving, Design of fast adders, Ripple carry adder and Carry look ahead adder, Multiplication of positive numbers, Problem Solving, Signed operand multiplication, Problem solving, Fast multiplication- Bit pair recoding of Multipliers, Problem Solving, Carry Save Addition of summands - Problem Solving, Integer division – Restoring Division - Solving Problems, Non-Restoring Division- Solving Problems, Floating point numbers and operations- Solving Problems.																

UNIT:3	(12 Hours)
Fundamental concepts of basic processing unit, Performing ALU operation, Execution of complete instruction, Branch instruction, Multiple bus organization, Hardwired control, Generation of control signals, Micro-programmed control, Microinstruction, Micro-program Sequencing, Micro instruction with Next address field, Basic concepts of pipelining, Pipeline Performance, Pipeline Hazards-Data hazards, Methods to overcome Data hazards, Instruction Hazards, Hazards on conditional and Unconditional Branching, Control hazards, Influence of hazards on instruction sets.	
UNIT:4	(12 Hours)
Parallelism, Need, types of Parallelism, applications of Parallelism, Parallelism in Software, Instruction level parallelism, Data level parallelism, Challenges in parallel processing, Architectures of Parallel Systems - Flynn's classification, SISD, SIMD, MIMD, MISD, Hardware multithreading, Coarse Grain parallelism, Fine Grain parallelism, Uni-processor and Multiprocessors, Multi-core processors, Memory in Multiprocessor Systems, Cache Coherency in Multiprocessor Systems, MESI protocol for Multiprocessor Systems.	
UNIT:5	(10 Hrs)
Memory systems -Basic Concepts, Memory hierarchy, Memory technologies, RAM, Semiconductor RAM, ROM, Types, Speed, size cost, Cache memory, Mapping Functions, Replacement Algorithms - Problem Solving, Virtual Memory, Performance considerations of various memories, Input Output Organization, Need for Input output devices, Memory mapped IO, Program controlled IO, Interrupts-Hardware, Enabling and Disabling Interrupts, Handling multiple Devices.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, "Computer Organization and Embedded Systems", Sixth Edition, Tata McGraw Hill, 2012. 2. David A. Patterson and John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface", Fourth Edition, Morgan Kaufmann / Elsevier, 2009. 3. Kai Hwang and F.A. Briggs, "Computer Architecture and Parallel Processing", McGraw Hill.	
Reference Books: 4. M. Morris Mano, "Computer System Architecture", PHI 5. William Stallings, "Computer Organization and Architecture – Designing for Performance", Sixth Edition, Pearson Education, 2003. 6. John P. Hayes, "Computer Architecture and Organization", Third Edition, Tata McGraw Hill, 1998. 7. John L. Hennessy and David A. Patterson, "Computer Architecture – A Quantitative Approach", Morgan Kaufmann / Elsevier Publishers, Fifth Edition, 2012.	

Subject Code	Name of the Subject	L	T	P	C										
	Design and Analysis of Algorithms	3	0	0	3										
Course Educational Objectives															
CEO1	Analyze the asymptotic performance of algorithms														
CEO2	Demonstrate familiarity with major algorithms														
CEO3	Apply important algorithmic design paradigms and methods of analysis														
CEO4	Synthesize efficient algorithms in common engineering design situations														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Apply efficient algorithms to reduce space and time complexity of both recurrent and														
CO2	Solve problems using divide and conquer approaches														
CO3	Apply greedy and dynamic programming types techniques to solve polynomial time														
CO4	Create exponential problems using backtracking and branch and bound approaches														
CO5	Interpret various approximation algorithms and interpret solutions to evaluate P type,														
CO6	Create algorithms that are efficient in space and time complexities by using divide														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	1	1												
CO2	2	2	2												
CO3	3	1	1												
CO4	3	1	1												
CO5	2	2	3												
CO6	1	3	2												
Avg.	2	1.6	1.6												
SYLLABUS															
Unit I (15Hrs)															
Introduction-Algorithm Design, Fundamentals of Algorithms, Correctness of algorithm, Time complexity analysis, Insertion sort-Line count, Operation count, Algorithm Design paradigms, Designing an algorithm, And its analysis-Best, Worst and Average case, Asymptotic notations Based on growth functions, $O, O(\Theta), \omega, \Omega$, Mathematical analysis, Induction, Recurrence relations- Solution of recurrence relations, Substitution method- Solution of recurrence relations, Recursion tree- Solution of recurrence relations.															
Unit -II (15Hrs)															
Introduction-Divide and Conquer, Maximum Subarray Problem, Binary Search, Complexity of binary search, Merge sort, Time complexity analysis, Quick sort and its Time complexity analysis, Best case, Worst case, Average case analysis, Strassen's Matrix multiplication and its recurrence relation, Time complexity analysis of Merge sort, Largest sub-array sum, Time complexity analysis of Largest subarray sum, Master Theorem Proof, Master theorem examples, Finding Maximum and Minimum in an array, Time complexity analysis-Examples, Algorithm for finding closest pair problem, Convex Hull problem.															
Unit – III (10Hrs)															
Introduction-Greedy and Dynamic Programming, Examples of problems that can be solved by using greedy and dynamic approach, Huffman coding using greedy approach, Comparison of brute force and Huffman method of encoding, Knapsack problem using greedy approach, Complexity derivation of knapsack using greedy, Tree traversals, Minimum spanning tree –															

greedy Kruskal's algorithm – greedy, Minimum spanning tree - Prims algorithm, Introduction to dynamic programming, 0/1 knapsack problem, Complexity calculation of knapsack problem, Matrix chain multiplication using dynamic programming, Complexity of matrix chain multiplication, Longest common subsequence using dynamic programming, Explanation of LCS with an example, Optimal binary search tree (OBST) using dynamic programming, Explanation of OBST with an example.	
Unit – IV	(10 Hrs)
Introduction to backtracking - branch and bound, N queen's problem – backtracking, Sum of subsets using backtracking, Complexity calculation of sum of subsets, Graph introduction, Hamiltonian circuit – backtracking, Branch and bound - Knapsack problem, Example and complexity calculation, Differentiate with dynamic and greedy, Travelling salesman problem using branch and bound, Travelling salesman problem using branch and bound example, Time complexity calculation with an example, Graph algorithms, Depth first search and Breadth first search, Shortest path introduction, Floyd-Warshall Introduction, Floyd-Warshall with sample graph, Floyd-Warshall complexity.	
Unit – V	(10 Hrs)
Introduction to randomization and approximation algorithm, Randomized hiring problem, Randomized quick sort, Complexity analysis, String matching algorithm – Examples, Rabin Karp algorithm for string matching - Example discussion, Approximation algorithm, Vertex covering, Introduction Complexity classes, P type problems, Introduction to NP type problems, Hamiltonian cycle problem, NP complete problem introduction, Satisfiability problem, NP hard problems – Examples.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book:	
1. <i>Introduction to Algorithms</i> , T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, 2nd Edition, PHI Learning Pvt. Ltd.	
2. <i>Fundamentals of Algorithm</i> , Horowitz & Sahani, 2nd Edition, Universities Press.	
Reference Book:	
1. <i>Algorithms, Design and Analysis</i> , H. Bhasin, First Edition, Oxford University	
2. <i>Design and Analysis of Algorithm</i> , S. Sridhar, Oxford University Press	
3. <i>Algorithms</i> , Sanjay Dasgupta, Umesh Vazirani, McGraw-Hill Education.	

Subject Code	Name of the Subject	L	T	P	C										
	Operating System	3	0	2	4										
Course Educational Objectives															
CE01	Provide an insight into the basic organization of computer systems.														
CE02	Familiarize students with various components of an Operating System														
CE03	Focus on fundamental problems and optimal solutions for resource management														
CE04	Understand different possible solutions for handling deadlock situation.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify the need of an Operating system														
CO2	Know the Process management functions of an Operating system														
CO3	Understand the need of Memory Management functions of an Operating system														
CO4	Find the significance of Device management role of an Operating system														
CO5	Recognize the essentials of File Management part of an Operating system														
CO6	Gain an insight of Importance of an Operating system through practical														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	2	2	1				1								
CO2	2	1	2				2								
CO3	1	1	3				3								
CO4	2	3	2				1								
CO5	3	1	1				1								
CO6	1	2	1				1								
Avg.	1.8	1.6	1.6				1.5								
SYLLABUS															
UNIT:1 (14 Hours)															
Operating System Objectives and functions, Gaining the role of Operating systems, The evolution of operating system, Major achievements, Understanding the evolution of Operating systems from early batch processing systems to modern complex systems, OS Design considerations for Multiprocessor and Multicore, Understanding the key design issues of Multiprocessor Operating systems and Multicore Operating systems, PROCESS CONCEPT– Processes, PCB, Understanding the Process concept and Maintenance of PCB by OS, Threads – Overview and its Benefits, Understanding the importance of threads, Process Scheduling : Scheduling Queues, Schedulers, Context switch, Understanding basics of Process scheduling, Operations on Process – Process creation, Process termination, Understanding the system calls – fork(),wait(),exit(),Inter Process communication : Shared Memory, Message Passing, Pipe(),Understanding the need for IPC, PROCESS SYNCHRONIZATION: Background, Critical section Problem, Understanding the race conditions and the need for the Process synchronization.															

UNIT:2	(12 Hours)
PROCESS SYNCHRONIZATION: Peterson's solution, Synchronization Hardware, Understanding the two-process solution and the benefits of the synchronization hardware, Process synchronization: Semaphores, usage, implementation, Gaining the knowledge of the usage of the semaphores for the Mutual exclusion mechanisms, Classical Problems of synchronization – Readers writers problem, Bounded Buffer problem, Good understanding of synchronization mechanisms, Classical Problems of synchronization – Dining Philosophers problem (Monitor) Understanding the synchronization of limited resources among multiple processes, CPU SCHEDULING: FCFS,SJF,Priority, Understanding the scheduling techniques, CPU Scheduling: Round robin, Multilevel queue Scheduling, Multilevel feedback Scheduling, Understanding the scheduling techniques, Real Time scheduling: Rate Monotonic Scheduling and Deadline Scheduling, Understanding the real time scheduling, DEADLOCKS: Necessary conditions, Resource allocation graph, Deadlock prevention methods, Understanding the deadlock scenario, Deadlocks :Deadlock Avoidance, Detection and Recovery, Understanding the deadlock avoidance, detection and recovery mechanisms	
UNIT:3	(14 Hours)
MEMORY MANAGEMENT: Memory Management: Logical Vs Physical address space, Swapping, Understanding the basics of Memory management, Contiguous Memory allocation – Fixed and Dynamic partition, getting to know about Partition memory management and issues: Internal fragmentation and external fragmentation problems, Strategies for selecting free holes in Dynamic partition, Understanding the allocation strategies with examples, Paged memory management, Understanding the Paging technique. PMT hardware mechanism, Structure of Page Map Table, Understanding the components of PMT, Example : Intel 32 bit and 64 –bit Architectures, Understanding the Paging in the Intel architectures, Example : ARM Architectures, Understanding the Paging with respect to ARM, Segmented memory management, Understanding the users view of memory with respect to the primary memory, Paged segmentation Technique, Understanding the combined scheme for efficient management	
UNIT:4	(14 Hours)
VIRTUAL MEMORY– Background, Understanding the need of demand paging, VIRTUAL MEMORY – Basic concepts – page fault handling, Understanding, how an OS handles the page faults, Performance of Demand paging, Understanding the relationship of effective access time and the page fault rate, Copy-on write, Understanding the need for Copy-on write, Page replacement Mechanisms: FIFO, Optimal, LRU and LRU approximation Techniques, Understanding the Pros and cons of the page replacement techniques, Counting based page replacement and Page Buffering Algorithms, To know on additional Techniques available for page replacement strategies, Allocation of Frames - Global Vs Local Allocation, Understanding the root cause of the Thrashing, Thrashing, Causes of Thrashing, Understanding the Thrashing, Working set Model, Understanding the working set model for controlling the Working set Model.	
UNIT:5	(14 Hours)
STORAGE MANAGEMENT: Mass storage structure – Overview of Mass storage structure – Magnetic Disks, Understanding the Basics in storage management, Disk Scheduling, Understanding the various scheduling with respect to the disk, FILE SYSTEM INTERFACE: File concept, File access methods, Understanding the file basics, File sharing and Protection,	

Emphasis the need for the file sharing and its protection, FILE SYSTEM IMPLEMENTATION: File system structure, To get the basic file system structure, Directory Implementation, Understanding the various levels of directory structure, FILE SYSTEM IMPLEMENTATION :Allocation methods, Understanding the pros and Cons of various disk allocation methods, FILE SYSTEM IMPLEMENTATION :Free space Management, Understanding the methods available for maintaining the free spaces in the disk, Swap space Management, Understanding the Low-level task of the OS.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books/References:

1. *Operating System Principles*, Abraham Silberchatz, Peter B. Galvin, Greg Gagne 8th Edition, Wiley Student Edition.
2. *Operating systems - Internals and Design Principles*, W. Stallings, 6th Edition, Pearson.M. G. Say, "Performance and design of AC machines", CBS Publishers, 2002.
3. *Modern Operating Systems*, Andrew S Tanenbaum 3rd Edition PHI.
4. *Operating Systems A concept - based Approach*, 2nd Edition, D. M. Dhamdhare, and TMH.
5. *Principles of Operating Systems*, B. L. Stuart, Cengage learning, India Edition.
6. *Operating Systems*, A. S. Godbole, 2nd Edition, TMH
7. *An Introduction to Operating Systems*, P.C.P. Bhatt, PHI.
8. *Operating Systems*, S, Haldar and A. A. Arvind, Pearson Education.
9. *Operating Systems*, R. Elmasri, A. G. Carrick and D. Levine, McGraw Hill.
10. *Operating Systems in depth*, T. W. Doeppner, Wiley.

Subject Code	Name of the Subject										L	T	P	C	QP
	Introduction to Machine Learning										3	0	0	3	A
Course Educational Objectives															
CEO1: Introduce Basics of Matrices, Random Variables and Distributions relevant for the study of Machine Learning Techniques															
CEO2: Formulate a well defined machine learning problem with clear Metrics															
CEO3: Familiarize with techniques for Dimensionality reduction and Computational Efficiency															
CEO4: Understand the notions of Hypotheses Space, Hypotheses Structure and Search															
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	Identify and classify different categories of Data attributes, Dimensions, Sample sizes														
CO2	Understand and Apply Supervised, Unsupervised Learning techniques														
CO3	Discriminate classifications based on Logistic and Linear Regression and Function Estimation														
CO4	Produce Rules and Associations for impactful recommendations from data														
CO5	Understand the applications of neural networks														
CO6	Apply the concept of reinforcement learning														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2													
CO2	3	2	2												
CO3	3	3													
CO4	2	3													
CO5	1														
CO6															
Avg.	2.4	2.5	2												
SYLLABUS															
UNIT:1 (12 Hours)															
Introduction to Machine Learning															
Introduction, Components of Learning, Learning Models, Geometric Models, Probabilistic Models, Logic Models, Grouping and Grading, Designing a Learning System, Types of Learning: Supervised, Unsupervised, Reinforcement, Perspectives and Issues, Version Spaces, PAC Learning, VC Dimension, Understand Your Data and Prepare Your Data for Analysis, Exploratory Data Analysis, Basic Probability Theory, Random Variables, Probability Distributions and Sampling Distributions, Markov Models, Bayesian Learning, Applications..															
UNIT:2 (10Hours)															
Supervised and Unsupervised Learning															
Decision Trees: ID3, Classification and Regression Trees, Regression: Linear Regression, Multiple Linear Regression, Logistic Regression, Support Vector Machines: Linear and Non-Linear, Kernel Functions, K Nearest Neighbors, Introduction to Clustering, K-means Clustering, K-Mode Clustering, Feature Engineering: Dimensionality Reduction: PCA, LDA, Feature Scaling, Feature Selection: Filter Methods, Wrapper Methods, Embedded Methods.															
UNIT:3 (12 Hours)															
Ensemble and Probabilistic Learning Model															
Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees															

Boosting: Adaboost, Stacking, Gaussian Mixture Models, The Expectation-Maximization (EM) Algorithm, Information Criteria, Nearest Neighbor Methods: Nearest Neighbor Smoothing, Efficient Distance Computations: KD-Tree, Distance Measures.	
UNIT:4	(14Hours)
Neural Networks Introduction, Early Models, Perceptron Learning, Backpropagation, Initialization, Training & Validation, Parameter Estimation: MLE, MAP, Bayesian Estimation, Unsupervised Learning, Partitional Clustering, Hierarchical Clustering, Density-Based Clustering, Gaussian Mixture Models, Expectation-Maximization, Introduction to Reinforcement and Explanation-Based Learning, ML Applications in Computer Vision, Recommendation Systems: Case Study Approach.	
UNIT:5	(12 Hours)
Reinforcement Learning: Introduction and Fundamentals, Key Concepts and Algorithms, Q-Learning, Deep Q-Networks, Policy Gradients, Applications of Reinforcement Learning, Advanced Topics in Machine Learning, Transfer Learning, Active Learning, Semi-Supervised Learning, Ethical Issues and Bias in Machine Learning, Future Trends in Machine Learning, Case Studies and Practical Implementations.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Machine Learning, <i>Tom M. Mitchell,(2013), McGraw-Hill Education (INDIAN EDITION)</i> 2. Hands-On Machine Learning with Scikit-Learn and TensorFlow, <i>AurélienGéron, (2017) O'Reilly</i> 3. Introduction to Machine Learning with Python A guide for data scientists, <i>Andreas, C. Muller & SarahGuido,(2016)O'Reilly</i>	
Reference Books: 1. <i>EthemAlpaydin, (2013), Introduction to Machine Learning, 2nd Ed., MIT Press.</i>	

V Semester [Third Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	HS	23BMGMG35001 / 23BMGMG35002	Engineering Economics and Costing/ Optimization Engineering	2	0	0	2	30	70
2	PC	23BCMPC35001	Software Engineering	3	0	0	3	30	70
3	PC	23BCMPC35002	Data Mining & Predictive Modelling	3	0	0	3	30	70
4	PE	23BCMPE350**	Professional Elective-III (From Bucket)	3	0	0	3	30	70
5	PC	23BCMPC35003	Advanced Machine Learning	3	0	0	3	30	70
6	OE	23BCMOE350**	Open Elective-I (From Bucket)	3	0	0	3	30	70
			TOTAL	17	0	0	17	180	420
PRACTICAL/SESSIONAL									
7	PC	23BCMPC35103	Advanced Machine Learning	0	0	2	1	30	20
8	PC	23BCMPC35102	Data Mining & Predictive Modelling	0	0	2	1	30	20
9	PST	23BCMPW35101	Minor Project -I	0	0	4	2	50	50
10	PST	23BCMPW35102	Summer Internship-II	0	0	1	1	0	50
11	MC	23BMCMC35001	Constitution of India	0	0	0	0	#100	
			TOTAL	0	0	9	5	110	140
			TOTAL	17	0	9	22	290	560

Professional Elective Bucket

Sl. No.	Sem	Professional Elective III
1	23BCMPE35011	Artificial Neural Network
2	23BCMPE35012	Genetic Algorithm and its Applications
3	23BCMPE35013	

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective I
1	23BCMOE35011	Advanced Java
2	23BCMOE35012	Web Technology
3	23BCMOE35013	Full Stack Development

Assignment Based Assessment

Subject Code	Name of the Subject										L	T	P	C	
	Software Engineering										3	0	0	3	
Course Educational Objectives															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Learn various software development process models and their suitability.														
CO2	Able to apply the methods of requirement elicitation.														
CO3	Learn Conventional Software Management and Evolution of Software Economics.														
CO4	Explain how a software process flows and understand Project Organizations and Responsibilities.														
CO5	Apply Agile methodologies and DevOps practices to enhance software development efficiency and collaboration														
CO6	Explore and integrate emerging trends and advanced techniques in software engineering for future-proof software solutions.														
CO-PO & PSO Mapping															
Cos	PROGRAMMEOUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1		3				2				1	1				
CO2		2		3											
CO3	1		1		2										
CO4		1													
Avg.	1	2	1	3	2	2				1	1				
SYLLABUS															
UNIT:1 (10 Hours)															
Introduction to Software Engineering, Software Process, Perspective and Specialized Process Models: The Waterfall Model, Incremental Process Models, The RAD Model, Prototyping Model, V- Model, Spiral Model, Agile and Scrum Model. COCOMO Model															
UNIT:2 (12 Hours)															
Software Requirements: Functional and Non-functional, User requirements, System requirements, Software Requirements Document, Requirement Engineering Process: Feasibility Studies, Requirement elicitation and analysis, Requirement validation, Requirements Management. CASE TOOL: Application on Documentation															
UNIT:3 (12 Hours)															
Design process: Design concepts, Design Model, Design Heuristic Architectural Design: Architectural Mapping using Data Flow Structure Chart design using DFD. Transform Analysis, Transaction Analysis: OOD Modeling Using UML, User Interface Design.															
UNIT:4 (14 Hours)															
Software implementation techniques: coding practices: Refactoring, Code Review, Code Inspection, Driver and Stub Module. Software Testing Fundamentals: Internal and external views of Testing- White Box Testing, Basis path testing, control structure testing. Black Box Testing: Regression Testing, Unit Testing, Integration Testing, Validation Testing, System Testing and Debugging. CASE TOOL: Test-Case Design, Software Reliability Measures and Growth Modeling: SEI CMM: Characteristics of Software Maintenance, Reverse Engineering, Re-engineering.)															
UNIT-V															
Emerging Trends and Advanced Topics in Software Engineering, Agile Software Development:															

Scrum, Kanban, Extreme Programming (XP), and Lean DevOps: Principles, Practices, and Tools , Cloud-Based Software Engineering and Microservices Architecture, Secure Software Development: Secure Coding Practices, Threat Modeling, and Vulnerability Analysis, Future Trends: AI in Software Engineering, Quantum Computing, and Green Software Engineering

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Book:

1. *Fundamentals of Software Engineering*, Rajib Mall, PHI, 2014.
2. *Walker Rayce : Software Project Management*, Pearson Education, 2005

Reference Book

1. *Software Engineering, A Practitioner's Approach*, Roger S. Pressman, TMG Hill.
2. *Software Engineering*, I. Sommerville, 9th Ed. , Pearson Education
3. *Richard H.Thayer: Software Engineering Project Management*, IEEE Computer Society, 1997
4. *ShereK.D. : Software Engineering and Management*, Prentice Hall, 1988

Subject Code		Name of the Subject								L	T	P	C	
		Data Mining and Predictive Modelling								3	0	0	3	
Course Educational Objectives														
CEO1	Teach students the leading trends and systems in data mining and predictive modelling													
CEO2	Make them understand the concepts of data mining, clustering, classification forecasting models, such that they are able to give the appropriate examples that will illustrate the above-mentioned concepts.													
CEO3	Teach them to develop different model and evaluate its performance													
CEO4	Enable students to be capable to describe the application based on predictive modelling and analysis.													
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>														
CO1	Demonstrate knowledge of various data mining techniques and predictive modelling algorithms.													
CO2	Apply data mining techniques for building model with pre-processing, implementing and evaluating.													
CO3	Develop skills in predictive modelling by applying the concepts of model training, validation, and evaluation to make accurate prediction													
CO4	Acknowledge time series analysis and its applications in predictive modelling													
CO5	Understand and analyse the ensemble methods													
CO6	Evaluate and validate the data mining models													
CO-PO & PSO Mapping														
COs	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1											1	
CO2	1	3	2		1						1	2	2	1
CO3	1	2	3	1	1						1		2	1
CO4	1	2	1								1	1	1	
CO5														
CO6														
Avg.	1.5	2	2	1	1						1	1.5	1.5	1
SYLLABUS														
UNIT:1 (10 Hours)														
Introduction to DMPM:														
Definition, Importance and Application of Data Mining, Introduction to Predictive Modeling, Data Pre-processing and Exploration Techniques, Exploratory Data Analysis: Descriptive Statistics and Data Visualization, Handling Missing Values and Outliers, Data Transformation and Normalization.														
UNIT:2 (12 Hours)														
Feature Selection and dimensionality reduction:														
Feature Selection Methods: Filter, Wrapper, Embedded Approaches, Principal Component Analysis, Linear Discriminant Analysis, Data Mining Techniques: Classification: Decision Tree														

Naïve Bayes, KNN.	
UNIT:3	(12 Hours)
Predictive Modelling Techniques: Clustering: K-means, Hierarchical Clustering, Association Rule Mining: Apriori Algorithm, Regression Analysis, Predictive Modeling Techniques: Regression Analysis: Linear Regression, Logistic Regression.	
UNIT:4	(12 Hours)
Ensemble Methods: Random Forests, Gradient Boosting, Support Vector Machine, Neural Networks, Association Rule Mining and Recommender Systems: Frequent Itemset Mining, Association Rule Generation and Evaluation, Collaborative and Content-Based Filtering.	
UNIT:5	(12 Hours)
Evaluation and Validation of Data Mining Models: Model Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, ROC Curves, Cross Validation Techniques, Overfitting and Underfitting, Time Series Analysis and Forecasting: Time Series Data Representation and Analysis, Smoothing Techniques: Moving Average, Exponential Smoothing, Forecasting Models: ARIMA, Neural Network.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1) Anasse Bari Mohamed Chaouchi, Tommy Jung, “Predictive Analysis for Dummies”, Wiley Publisher, 2nd edition, 2016 2) Bertt Lantz, Machine Learning with R: Expert Techniques for Predictive modelling to solve all your data analysis problems, Pack Publisher, 2nd Edition, 2015. 3) Data Mining: Concepts and techniques: J Han and M Camber , Elsevier.	
Reference Books: 1) Aurelien, “Hands-On Machine Learning with Scikit-Learn & TensorFlow”, O’Reilly Publisher, 5th Edition, 2017. 2) Max Kuhm, Kjell Jhonson, “ Applied Predictive Modeling”, Springer, 2013. 3) Data Mining- A Tutorial based Primer by R.J roiger, M.W. geatz, Pearson Education.	

			Advanced Machine Learning					3	0	0	3			
Course Educational Objectives														
CEO1	Master Complex Machine Learning Algorithms													
CEO2	Develop and Optimize Predictive Models													
CEO3	Handle and Transform Complex Data													
CEO4	Address Ethical and Practical Challenges													
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>														
CO1	Describe and explain advanced machine learning algorithms such as ensemble methods, support vector machines, and deep learning architectures.													
CO2	Optimize models through techniques like hyperparameter tuning and regularization to improve their performance.													
CO3	Critically assess the impact of preprocessing on model performance and data quality.													
CO4	Assess the suitability of reinforcement learning and deep learning approaches for various real-world applications.													
CO5	Integrate machine learning models into production environments, addressing challenges related to system deployment, real-time performance, and scalability.													
CO6	Examine the ethical implications of machine learning technologies, including issues related to bias, fairness, and privacy.													
CO-PO & PSO Mapping														
COs	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1											1	
CO2	1	3	2		1						1	2	2	1
CO3	1	2	3	1	1						1		2	1
CO4	1	2	1								1	1		
CO5	1	3	2		1						1	2		
CO6	1	2	3	1	1						1			
Avg.	1.3	2.1	2.2	1	1						1	1.6	1.6	1.3
SYLLABUS														
UNIT:1 (10 Hours)														
Introduction to Machine Learning, Type of Machine Learning Architecture, Type of Neural Network, Learning Rate, Activation Function, Supervision Learning: Classification methods advantages and disadvantages, Implementation using Python. Regression methods advantages and disadvantages, Implementation using Python.														
UNIT:2 (12 Hours)														
Supervised Learning: Classification methods, Advantages and Disadvantages, Implementation using Python, Regression methods, Advantages and Disadvantages, Implementation using Python.														
UNIT:3 (12 Hours)														
Unsupervised Learning: Clustering methods, Pros and Cons of different clustering algorithms, Implementation using Python.														
UNIT:4 (12 Hours)														

Reinforcement Learning: Introduction and its Types, Implementation using Python, Ensemble Learning: Introduction, Types of standard ensemble learning.

UNIT:5

(12 Hours)

Pros and Cons of Ensemble Learning, Implementation using Python, Performance parameters of Classification, Regression, and Clustering, Model validation and its types, Introduction to Hyper-parameters and their Types, Methods of Hyper-parameter Tuning, Machine Learning Life Cycle.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

- 1) Christopher M. Bishop, Pattern Recognition and Machine Learning, 1st Edition, 2006
- 2) Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 1st Edition, 2012

Reference Books:

- 1) Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer, 2nd Edition, 2009.
- 2) Richard S. Sutton, Andrew G. Barto, Reinforcement Learning: An Introduction, MIT Press, 2nd Edition, 2018.

Subject Code	SUBJECT NAME											L	T	P	C
	Artificial Neural Networks												4	0	3
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO 01	CO 01: Understand the components and functions of biological and artificial neurons and neural networks.														
CO 02	Apply different activation and propagation functions and learning strategies in neural networks.														
CO 03	Develop and implement supervised learning models like perceptron, back propagation, multilayer perceptron, and radial basis function networks.														
CO 04	Design and train advanced neural network models such as recurrent networks and self-organizing maps.														
CO 05	Assess the performance of neural network models using various metrics and techniques, and optimize models through regularization, hyperparameter tuning, and cross-validation.														
CO 06	Implement artificial neural networks using modern machine learning frameworks and libraries, such as TensorFlow or PyTorch, and apply them to real-world datasets.														
CO-PO & PSO Mapping (Articulation Matrix):															
COs	PROGRAMME OUTCOMES												PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02	
01	3														
02	2														
03	3	2													
04	2			3											
05		2		3											
06		2		3											
Avg.	2.5	2		3											
Justification of the Mapping:															
CO	PO	Level	Justification												
01	01	3	Understanding the motivation of ANN from Biological neuron is engineering acquiring engineering knowledge												
02	02	2	Solve the problem with input neuron, weights and bias												
03	01	3	Learning about Perceptron is Engineering Knowledge												
03	02	2	Problem Analysis of neural network with Activation functions.												
04	01	1	Engineering Knowledge												
04	04	3	Investigation of complex problems of RNN.												
Syllabus															
UNIT I												10 Hours			
Why neural network?, Basics of Artificial Neural Networks, A brief history of neural networks, Biological neural networks, The vertebrate nervous system: peripheral nervous system, Cerebrum, cerebellum, diencephalon, brainstem, The Neuron, Components of Neuron, Electrochemical processes, Receptor cells - Various types, Information processing with the nervous system, Light Sensing organs, Neurons in living organisms, Transitions in technical neurons..															
UNIT II												10 Hours			
Components of artificial neural networks, The concept of time in neural networks, Connections,															

Propagation function, Activation, Threshold value, Activation function, Common activation functions, Output function, Learning strategies, Network topologies: Feed forward networks, Recurrent Networks, Completely linked networks, Bias neuron, Representing Neurons, Orders of activation: Synchronous activation, Asynchronous activation, Input and output of data.		
UNIT III		12 Hours
Learning and training samples, Paradigms of learning, Using training samples, Gradient Optimization Procedure, Hebbian learning rule, Supervised learning network paradigms, The perceptron, Single layer perceptron, Linear Separability, Multilayer perceptron, Back propagation of error, Selecting learning rate, Resilient Back propagation, Adaptation of weights, Variations in Back propagation, Multilayer perceptron.		
UNIT IV		12 Hours
Radial basis functions, Information processing of an RBF network, Training of RBF networks, Growing of RBF networks, Compare multilayer perceptrons and RBF, Recurrent perceptron-like networks, Jordan Networks, Elman networks, Training recurrent networks, Unfolding in time, Teacher forcing, Recurrent back propagation, Evolutionary algorithms.		
UNIT V		12 hours
Unsupervised learning network paradigms, Structure of a self-organizing map (SOM), Functionality, Training, Topology function, Decreasing Learning Rate, Variations of SOMs, Neural Gas, Multi-SOM, Multi-Neural Gas, Growing neural gas, Adaptive resonance theory (ART), Task and structure of an ART network, Resonance, Learning process of an ART network.		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books: David Kriesel, A Brief Introduction to Neural Networks, dkriesel.com, 2005. Gunjan Goswami, Introduction to Artificial Neural Networks, S.K. Kataria & Sons, 2012.		
Reference Book: Raul Rojas, Neural Networks: A Systematic Introduction, 1996. S. Sivanandam, Introduction to Artificial Neural Networks, 2003		

	Advanced Java Programming										4	0	0	4
Course Educational Objectives														
CEO1	Objective of this course is to provide the ability to design console based, GUI based and web based applications. Students will also be able to understand integrated development environment to create, debug and run multi-tier and enterprise-level applications.													
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>														
CO1	design and develop data driven web applications using JDBC, servlet and JSP API.													
CO2	develop MVC based web applications using SpringBoot.													
CO3	understand significance of Object Relational Mapping (ORM) in Java context.													
CO4	construct ORM-based web applications using Hibernate.													
CO5	Develop proficiency in advanced Java programming concepts.													
CO6	Utilize Java's networking and I/O capabilities to build robust and scalable													
CO-PO & PSO Mapping (Articulation Matrix):														
Cos	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1	2	3				3						
CO2		3	2					3						
CO3	2													
CO4		2	2											
CO 6			2											
CO 7														
Avg.	2.5	2	2					3						
SYLLABUS														
UNIT:1 (12 Hours)														
Basics of JDBC: Introduction to JDBC, Need of JDBC, JDBC Drivers (4 types), Architecture of JDBC, Components of JDBC(Classess and Interfaces).														
Programming with JDBC: Creating a DATABASE (ORACLE), First Program to connect to the DATABASE created, Loading the Driver, Establishing the Connection, Creating Statements (Statement/ Prepared Statement/ Callable Statement), Executing a SQL Query, Different types of SQL Queries.														
UNIT:2 (12 Hours)														
JDBC Program to Retrieve Data from DATABASE: Introduction to Result Set, Result Set with Statement Interface, Result Set with Prepared Statement Interface, Updating and deleting data in database, Result Set Metadata, Executing Stored Procedures Using Callable Statement.														
Web Architecture: Introduction to HTML, Describe the various HTML tags and their usage, Layout Tags, Semantic Tags, HTML table, forms, style, div.														
UNIT:3 (10 Hours)														
Introduction to Servlets: What is Servlet, Advantage of Servlet Over Applets and CGI, Strengths of Servlet, Architecture of Web Application, Web Servers and its Containers, Role of servlet in Web application development, Understanding servlet-api, Understanding HTTP protocol and communication between HTML-SERVLET.														
Getting Deep into Servlets: Types of Servlet, Difference between HttpServlet and GenericServlet, Life cycle of Servlets and different life cycle methods, Difference between doGet() and doPost(), Servlet Generating HTML output, Collecting Client submitted data in a Servlet.														
UNIT:4 (14 Hours)														
Servlet Communications: Servlet to DBMS communication using type-4 connection, Servlet to DBMS														

communication using JDBC connection pooling, Servlet communication with other servlets (Servlet Chaining), Servlet communication with JSP or HTML page (sendRedirect(), Difference between sendRedirect() and RequestDispatcher forward(), Understanding ServletConfig. Conclusion to Servlets: Servlet Filters and Wrappers, Servlet Listeners, Session Tracking, Cookies, HttpSession, HTML hidden form field element, URL rewriting, Annotation-based servlet programs..

UNIT:5

(14 Hours)

Java Server Pages (JSP): Introduction to JSP, Scope of JSP, Anatomy of a JSP program, Execution of a JSP program, Significance of JSP Engine, Built-in objects of JSP, Significance of JSP Elements, Scripting Elements, Scriptlets, Declaration, Expression, Directives and Action Elements, Page Directive, Include Directive, Taglib Directive, Forward action element, Include, Param, useBean with introduction to beans, setProperty, getProperty. Hibernate: Introduction to Hibernate, Architecture of Hibernate, Hibernate O/R Mapping, Configuring Hibernate development environment, Implementing Hibernate O/R Mapping, Introduction to Hibernate Query Language (HQL). Spring and Spring Boot: Spring Basics, Spring IOC, Spring MVC, Spring MVC with Hibernate, Introduction to Spring Boot, Spring Boot Injection, Dependency, Autowired, Spring Boot JPA, Spring Boot Webapp MVC.

Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs

Text Books:

1. Advanced Java Programming, Uttam K. Roy, Oxford University Press.

Reference Book:

1. Black book, Kogent Learning Solution Inc.
2. Java 2: The Complete Reference by Herbert Schildt, Fifth Edition Paperback

VI Semester [Third Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

S N	CC	Sub Code	Name of the Subject	L	T	P	C	IS A	ES A
THEORY									
1	HS		Engineering Economics and Costing/ Optimization Engineering	2	0	0	2	30	70
2	PC	23BCMPC3600 1	Computer Networks	3	0	0	3	30	70
3	PC	23BCMPC3600 2	Application of Machine Learning in Industries	3	0	0	3	30	70
4	PE	23BCMPE360**	Professional Elective-IV (From Bucket)	3	0	0	3	30	70
5	PC	23BCMPC3600 3	Microprocessors and Microcontrollers	3	0	0	3	30	70
6	OE	23BCMOE360**	Open Elective-II (From Bucket)	3	0	0	3	30	70
			TOTAL	17	0	0	17	180	420
PRACTICAL/SESSIONAL									
7	PC	23BCMPC3610 1	Computer Networks Lab	0	0	2	1	30	20
8	PC	23BCMPC3610 2	Deep Learning using python	0	0	2	1	30	20
9	PS T	23BCMPW3610 1	Minor Project Work-II	0	0	4	2	50	50
10	MC		Essence of Indian Knowledge Tradition	0	0	0	0	#100	
			TOTAL	0	0	8	4	110	90
			TOTAL	17	0	8	21	290	510

Professional Elective Bucket

Sl. No.	Sem	Professional Elective IV
1	23BCMPE36011	Deep Learning
2	23BCMPE36012	Intelligent Machining
3	23BCMPE36013	High Performance Computing

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective II
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to Analog Signal Conversion, Analog to Analog Signal Conversion, Analogue to Digital Signal conversion. Digital Transmission types (Serial and Parallel), Transmission Modes (Simplex, Half-Duplex and Full-Duplex), Multiplexing (FDM, TDM, WDM, CDM).	
UNIT:2	(10 Hours)
Switching and Types: Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.	
Data link layer: Introduction, Error Detection and Correction techniques (Checksum, Cyclic Redundancy Check, Hamming Code (1-bit error rectification)), Framing, HDLC Frame Structure and Types, HDLC (NRM, ABM, ARM), Point to Point Protocol, Flow and Error Control, Noiseless Channels protocols (Simplest, stop-and-Wait), Noisy Channels protocols (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ),	
UNIT:3	(11 Hours)
Medium Access sub layer: Multiple Access - Random Access Protocols (ALOHA, CSMA (non-persistent, 1-persistent, p-persistent), CSMA/CD, CSMA/CA), Controlled Access Protocols (Reservation, Polling, Token Passing), Channelization techniques (FDMA, TDMA, CDMA).	
UNIT:4	(10 Hours)
Network Layer: Internetworking, IP Address, IPv4(Header structure, Class types, subnetting), IPv6 (Header structure, Address structure), IPV4 to IPV6 Address Mapping, Address Resolution Protocol, ICMP.	
UNIT:5	(10 Hours)
Transport Layer: User Datagram Protocol, Transmission Control Protocol, Process-To-Process Delivery, Congestion Control, Quality of Service.	
Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book:	
1. <i>Data Communications and Networking – Behrouz A. Forouzan, Fourth Edition TMH,2006.</i> 2. <i>Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education.</i>	
Reference Book:	
1. <i>An Engineering Approach to Computer Networks-S.Keshav,2nd Edition, Pearson Education</i> 2. <i>Understanding communications and Networks,3rd Edition, W.A.Shay, Cengage Learning.</i> 3. <i>Computer and Communication Networks ,Nader F. Mir, Pearson Education</i> 4. <i>Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose, K.W. Ross, 3rd Edition, Pearson Education.</i>	

Subject Code	Application of Machine Learning in Industry											L	T	P	C
												4	0	0	
Course Educational Objectives															
CEO 01	Explain the concepts of IoT and Industry 4.0.														
CEO 02	Process of implementing the technologies of IoT in industry.														
CEO 03	Impart necessary and practical knowledge of components of Internet of Things and develop skills required to build real-life IoT based projects.														
CEO 04	Impart necessary and practical knowledge of components of Internet of Things in industry and develop skills required to build real-life IoT based projects.														
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO 01	Understand internet of Things and its hardware and software components.														
CO 02	Interface I/O devices, sensors & communication modules.														
CO 03	Understanding the concept of Industry 4.0														
CO 04	Develop real life Industry 4.0 and Machine Learning based projects.														
CO 05	Understand and articulate the role of machine learning in industrial contexts, including its impact on productivity, efficiency, and decision-making processes.														
CO 06	Develop and implement machine learning models to solve industry-specific problems, such as predictive maintenance, quality control, and supply chain optimization.														
CO-PO & PSO Mapping (Articulation Matrix):															
COs	PROGRAMME OUTCOMES												PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02	
01	3	1			1							1	1	-	
02	2	3	1	3	2							1	1	1	
03	2	3	3	3	2				1		1	1	1	1	
04	2	3	3	3	2				1		1	1	1	1	
05	2	3	3										1	1	
06	2	3	3										1	1	
Avg.	2.25	2.6	2.6	3	1.75				1		1	1.0	1.0	1.0	
Syllabus															
UNIT I													8 Hours		
Introduction to IoT: Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals - Devices and gateways, Data management, Business processes in IoT, Everything as a Service (XaaS), Role of Cloud in IoT, Security aspects in IoT. Domain-Specific IoTs: Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health & Life Style.															
UNIT II													09 Hours		
Elements of IoT: - Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for Communication Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.															
UNIT III													18 Hours		

Introduction to Industry 4.0: -Introduction to Industry 4.0: Overview of Industry 4.0 and its key principles, Evolution of manufacturing and the role of automation, Integration of cyber-physical systems in Industry 4.0. Predictive Maintenance: Understanding the importance of predictive maintenance in Industry 4.0. Case Studies and Real-World Applications: Quality Control and Inspection - ML applications in quality control processes, Image recognition and computer vision for defect detection, Practical applications and case studies, Hands-on exercises on predictive maintenance algorithms.		
UNIT IV		10 Hours
Supply Chain Optimization: - Supply Chain Optimization: Role of ML in optimizing supply chain operations, Demand forecasting and inventory management, Logistics planning and route optimization. Digital Twins and Simulation: Concept of digital twins in Industry 4.0, Simulation and modeling for process optimization, Building and using digital twins in manufacturing.		
UNIT V		10 Hours
Human-Robot Collaboration: ML-enabled collaboration between humans and robots, Safety considerations and case studies.		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books: 4. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014. 5. VijayMadiseti, Arshdeep Bahga," Internet of ThingsA Hands- On- Approach",2014,ISBN:978 0996025515 6. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs 7. "Introduction to Industry 4.0: How to Revolutionize Your Business with Digitalization" by Peter Zec 8. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron 9. Research papers and case studies on specific Industry 4.0 applications		
Reference Book: 5. AdrianMcEwen, "Designing the Internet of Things", Wiley Publishers, 2013, ISBN:978-1-118-43062-0 6. Daniel Kellmereit, "The Silent Intelligence: The Internet of Things". 2013, ISBN:0989973700 7. Adrian McEwen, "Designing the Internet of Things", Wiley 8. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill 9. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media		

Subject Code	Name of the Subject										L	T	P	C	
BPECS7031	DEEP LEARNING										3	0	0	3	
Course Educational Objectives															
CEO1	Understand the concepts of Neural Networks and Deep Learning.														
CEO2	Understand Deep neural network and layered learning approach.														
CEO3	Study and understand CNN and RNN for deep learning.														
CEO4	Learn and understand Auto Encoders and its applications and Understand concept of transfer learning and its applications with keras.														
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	Apply basic mathematical concepts in Deep Learning.														
CO2	Work with powerful framework for supervised learning.														
CO3	Deal with Convolution Neural Networks.														
CO4	Analyze various types efficient data encoders, Apply various network models in deep learning.														
CO 5	Design, implement, and evaluate deep learning models for various applications, including image recognition, natural language processing, and time-series forecasting, using popular frameworks like TensorFlow and PyTorch.														
CO 6	Understand and apply advanced deep learning techniques, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and autoencoders, to solve complex problems and optimize model performance.														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	3	3		3							3	3	3	0
CO2	3	3			3							3	3	3	2
CO3	3	3	3		3							3	3	3	3
CO4	3	3		3	3							3	3	3	3
CO 5	3	3													
CO 6		3	3												
Avg.	3	3	3	1	3							3	3	3	2
SYLLABUS															
UNIT:1 (14 Hours)															
Historical trends in deep learning – Machine Learning basics, Learning algorithms – Supervised and Unsupervised Training, Linear Algebra for machine learning, Testing - Cross Validation, Dimensionality Reduction ,Over fitting/Under fitting, Hyper parameters and validation sets, Estimators – Bias – Variance, Loss Function—Regularization, Biological Neuron – Idea of Computational units, McCulloch-Pitts units and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Convergence theorem for Perceptron, Learning Algorithm, Linear Separability, Multilayer perceptron –The first example of network with Keras code, Back propagation.															
UNIT:2 (14 Hours)															
Introduction to Simple DNN, Platform for Deep Learning, Deep Learning Software Libraries, Deep Feed Forward Networks Introduction, Learning XOR, Gradient-based Learning, Various															

Activation Functions, ReLU, Sigmoid – Error Functions, Architecture Design, Differentiation Algorithms, Regularization methods for Deep Learning, Early Stopping, Drop Out, Difficulty of training deep neural networks, Greedy layer wise training, Optimization methods for Neural Networks-Adagrad, Adam.	
UNIT:3	(14 Hours)
Convolution Neural Networks Introduction, Convolution Operation, Motivation, Pooling, Normalization, Applications in Computer Vision – ImageNet, Sequence Modelling –VGGNet, LeNet, Recurrent Neural Networks, RNN topologies- Difficulty in Training RNN, Long Short Term Memory, Bidirectional LSTMs, Bidirectional RNNs, Application case study -Handwritten digits recognition using deep learning, LSTM with Keras – sentiment Analysis.	
UNIT:4	(14 Hours)
Encoder, Decoder, Auto Encoders Introduction: Auto Encoders, Under Complete Auto Encoders, Regularized Auto Encoders, Stochastic Auto Encoder, Denoising Auto Encoder, Contractive Auto Encoder, Auto Encoder Applications. Dimensionality Reduction and Classification using Auto Encoders: Recommendation.	
UNIT:5	
Optimization for Deep Learning: Optimizers - RMSprop for RNNs, SGD for CNNs. Application Case Study: Image Dimensionality Reduction Using Encoders. LSTM with Keras: Sentiment Analysis.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Ian Good Fellow, Yoshua Bengio, Aaron Courville,“Deep Learning”, MIT Press,2016. 2. Kevin P. Murphy, “Machine Learning: A Probabilistic Perspective”, MIT Press, 2012.	
Reference Books: 1. <i>Neural Networks: A Systematic Introduction</i> , RaulRojas, 1996. 2. <i>Christopher and M. Bishop ,”Pattern Recognition and Machine Learning”, Springer Science Business Media, 2006.</i> 3. <i>Jason Brownlee,” Deep Learning with Python”, ebook, 2016.</i>	

Subject Code		Name of the Subject						L	T	P	C	QP			
BOECS6040		Android App Development Using Kotlin						4	0	1	4	A			
Course Educational Objectives															
CEO1	This course covers the fundamentals of Kotlin Programming and Android programming using the Android SDK. Topics discussed in this course include: fundamental concepts in Android programming - activities and intents, designing user interface using views, data persistence, content providers, messaging and networking, location-based services, and developing android services.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	demonstrate their understanding of Kotlin Programming.														
CO2	demonstrate their understanding of the fundamentals of Android operating systems, development trends and Android platform.														
CO3	Design kotlin based stand android application.														
CO4	Develop android application using API and database.														
CO5	Design, develop, and deploy functional Android applications using core Android components such as Activities, Services, Broadcast Receivers, and Content Providers, while employing best practices for UI/UX design and adhering to Android architecture guidelines.														
CO6	Implement advanced features in Android applications, including dynamic UI elements, complex data handling with XML parsing, notifications, and interactions with various Android components (e.g., Intents, Widgets), using Kotlin or Java to enhance application functionality and performance.														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1	3				2							1			
CO2	2	3	1	3	2							1			
CO3	2	3	3	3	2				1		1	2			
CO4	2	3	3	3	2				1		1	2			
CO 5	1	3													
CO 6			3	1											
Avg.	2	3	2.5	2.5	2				1		1	1.5			
SYLLABUS															
UNIT:1 (8 Hours)															
Introduction to Kotlin: Kotlin Architecture, Basic syntax, Kotlin Comments, Kotlin Keywords, Defining variable, Use of variables, Data types- (Booleans, Strings, Arrays, Ranges), Operators, Control Flow, if... else expression, for loops, While loops, Break and Continue, functions in Kotlin, user defining functions. Kotlin Collections (Lists, Sets, Maps), Kotlin objects and Classes, Constructors, Inheritance, Interface, Visibility Control, Extension, Data Classes, Sealed Class, Generics, Delegation, Destruction, Exception Handling.															
UNIT:2 (15 Hours)															
Introduction to Android: Introduction to Android Operating System, Android Architecture, Core Building Blocks, Android Emulator, Application Structure, Setting up Development															

Environment, Dalvik Virtual Machine & .apk File Extension, AndroidManifest.xml, Uses-Permission & Uses-SDK, Resources & R.java, Assets. Fundamentals: Basic Building Blocks - Activities, Services, Broadcast Receivers & Content Providers, UI Components - Views & Notifications, Components for Communication - Intents & Intent Filters, Android API Levels (Versions & Version Names). User Interface Architecture: Application Context, Intents, Activity Lifecycle, Services, Android Widgets, UI Widgets, Working with Buttons (Button, Toast, Custom Toast, Toggle Button, Check Button, Custom Check Button, Radio Button, Dynamic Radio Button, Alert Dialog). Android Views: TextView, Autocomplete TextView, ListView, Custom ListView, Rating Bar, WebView, Seek Bar, Date Picker, Time Picker, Analog and Digital, Progress Bar	
UNIT:3	(10 Hours)
Android Layouts: Linear Layout, Relative Layout, Table Layout, Table Layout with Frame Layout, Search View, Search View on Toolbar, EditText with Text Watcher. Switchers: Switch, Dynamic Switch, Text Switcher, Dynamic Text Switcher, Image Switcher, Dynamic Image Switcher. Spinner: Spinner, Dynamic Spinner, Dynamic TextClock, Chronometer, Dynamic Chronometer. Notifications and Other Components: Notifications, Android Toast, Dynamic Radio Group, Android Slide Up/Down in Kotlin, Scroller (Vertical Scroll, Horizontal Scroll View), Image Slider, View Stub. XML Parsing: XML Parsing Using DOM Parser, XML Parsing Using SAX Parser, Kotlin Android XML PullParser.	
UNIT:4	(10 Hours)
Menu: Option menu, Context menu, Sub menu, menu from xml, menu via code, Examples. Navigation: Create a fragment, define navigation paths, start an external activity, Kotlin Regex: Regular Expressions, Kotlin Annotations, Adapters, Android RecyclerView, Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources.	
UNIT: 5	
Data Storage: Shared Preferences, Android File System, Internal storage, External storage, SQLite Firestore. Common Android APIs: Using Android Data and Storage APIs etc.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Bruce Eckel and Svetlana Isakova (2021), “Atomic Kotlin”, Mindview LLC, Crested Butte, CO 2. WeiMeng Lee (2012), “Beginning Android Application Development”, Wrox Publications (John Wiley, New York) (For 1 to 5 units).	
Reference Books: 1. Ed Burnette (2010), “Hello Android: Introducing Google's Mobile Development Platform”, The Pragmatic Publishers, 3rd edition, North Carolina USA 2. Reto Meier (2012), “Professional Android 4 Application Development”, Wrox Publications (John Wiley, New York). 3. ZigurdMednieks, Laird Dornin, Blake Meike G, Masumi Nakamura (2011), “Programming Android: Java Programming for the New Generation of Mobile Devices”, O'Reilly Media, USA	

e-Contents:

1. <https://developer.android.com/kotlin>
2. <https://developer.android.com/studio>

VII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

S N	CC	Sub Code	Name of the Subject	L	T	P	C	IS A	ES A
THEORY									
1	PC	23BCMPC4700 1	Computer Graphics	3	0	0	3	40	60
2	PC	23BCMPC4700 2	Formal Language & Automata Theory	3	0	0	3	40	60
3	PE	23BCMPE470**	Professional Elective-V (From Bucket)	3	0	0	3	40	60
4	HS	23BHS4700 1	Entrepreneurship Development	3	0	0	3	40	60
5	OE	23BCM4700**	Open Elective-III (From Bucket)	3	0	0	3	40	60
			TOTAL	15	0	0	15	200	300
PRACTICAL/SESSIONAL									
5	PC			0	0	2	1	70	30
5	PS T	23BCMPW47001	Major Project Work-I	0	0	6	3	50	50
6	PS T	23BCMPW47001 2	Summer Industry Internship-III	0	0	0	1	0	50
			TOTAL	0	0	8	5	120	80
			TOTAL	15	0	8	20	320	380

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Sem	Professional Elective V
1	23BCMPE47011	Computational Linguistics & NLP
2	23BCMP12E47012	Robotics: Computational motion planning
	23BCMPE47013	Advanced Datamining

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective III
1	23BCM47011	Computer Vision
2	23BCM47012	Software Project Management
	23BCM47013	Cryptography and Network

UNIT:2	(12 Hours)
Two Dimensional Geometric Transformations: Translation, rotation, Scaling, Reflection, Shear, Matrix Representation, Composite Transformations, Transformation between coordinate systems. Window-to- View port Coordinate Transformation. Line Clipping (Cohen-Sutherland Algorithm) and Polygon Clipping (Sutherland Hodgeman Algorithm). Three Dimensional Geometric and Modeling Transformations: Translation Rotation, Scaling, Reflections, shear, Composite Transformation. Projections: Parallel Projection and Perspective Projection.	
UNIT:3	(10 Hours)
Object representation: Spline Representation, Bezier Curves and B-Spline Curves. Fractal Geometry: Fractal Classification and Fractal Dimension. Visible Surface Detection Methods: Back-face Detection, Depth Buffer, A- Buffer, Scan- line Algorithm and Painters Algorithm.	
UNIT:4	(12 Hours)
Light pattern and Illumination Models: Aliasing and Anti aliasing, Half toning, Thresholding and Dithering, Basic Models, Displaying Light Intensities. Surface Rendering Methods: Polygon Rendering Methods: Gouraud Shading and Phong Shading.	
UNIT-5	[10 Hours]
Computer Animation: Types of Animation, Key frame Vs. Procedural Animation, methods of controlling Animation, Morphing. Virtual Reality: Types of Virtual reality systems, Input and Output Virtual Reality devices	
Teaching Methods: Chalk& Board/ PPT	
Text Books: 1. <i>Computer Graphics with Virtual Reality System, Rajesh K.Maurya, WileyDreamtech.</i> 2. <i>Computer Graphics, D. Hearn and M.P. Baker (C Version), Pearson Education</i>	
Reference Book: 1. <i>Computer Graphics Principle and Practice , J.D. Foley, A.Dam, S.K. Feiner, Addison, Wesley</i> 2. <i>Procedural Elements of Computer Graphics- David Rogers (TMH)</i> 3. <i>Computer Graphics: Algorithms and Implementations – D.P Mukherjee &Debasish Jana (PHI)</i> 4. <i>Introduction to Computer Graphics & Multimedia – AnirbanMukhopadhyay</i>	

Subject Code		Name of the Subject								L	T	P	C			
BPECS7014		Natural Language Processing								3	0	0	3			
Course Educational Objectives																
CEO1	Teach students the leading trends and systems in natural language processing.															
CEO2	Make them understand the concepts of morphology, syntax, semantics and pragmatics of the language and that they are able to give the appropriate examples that will illustrate the above mentioned concepts.															
CEO3	Teach them to recognize the significance of pragmatics for natural language understanding.															
CEO4	Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Understand approaches to syntax and semantics in NLP.															
CO2	Understand approaches to discourse, generation, dialogue and summarization within NLP.															
CO3	Understand current methods for statistical approaches to machine translation.															
CO4	Understand machine learning techniques used in NLP, including the probabilistic context-free grammars and unsupervised methods, as applied within NLP.															
CO 5	Apply various NLP techniques and algorithms for tasks such as information extraction, semantic search, and text summarization, including both extractive and abstractive methods, to process and analyze natural language data effectively.															
CO 6	Develop and implement advanced NLP applications, such as chatbots and machine translation systems, by leveraging natural language understanding (NLU) and natural language generation (NLG) techniques, and evaluate their performance based on real-world use cases.															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	2	3	3	3	-	-	-	3	2	2	3	3	3	3	
CO2	3	2	3	3	3	-	-	-	3	2	2	3	3	3	3	
CO3	3	0	3	3	3	-	-	-	3	2	2	3	3	3	3	
CO4	3	2	3	3	3	-	-	-	3	2	2	3	3	3	3	
CO5	3	3											3	3	3	
CO6	3	3											3	3	3	
Avg.	3	2	3	3	3				3	2	2	3	3	3	3	
SYLLABUS																

UNIT:1 (12 Hours)	
Introduction to Natural Language Processing, Steps – Morphology – Syntax –Semantics, Morphological Analysis (Morphological Parsing), Stemming – Lemmatization, Parts of Speech Tagging, Approaches on NLP Tasks (Rule-based, Statistical, Machine Learning),N-grams, Multiword Expressions, Collocations (Association Measures, Coefficients and Context Measures),Vector Representation of Words ,Language Modeling.	
UNIT:2 (12 Hours)	
Syntax Parsing, Dependency Parsing, Semantics, Semantic Parsing, Word Sense Disambiguation, Lexical Disambiguation, Structural Disambiguation, Word, Context and Sentence-level Semantics, Pronoun Resolution, Semantic Representation of text, Introduction to Semantic Relations.	
UNIT:3 (10 Hours)	
Semantic Relations, Semantic Role Labeling, Semantic Frames, Ontology and Semantics, Semantic Network and Knowledge Graph, Intent Detection and Classification, Paraphrase Extraction, Discourse, Coreference Resolution, Text Coherence, Discourse Structure, Coherence Discourse Planning	
UNIT:4 (12 Hours)	
Information Extraction and Its Approaches: Information Extraction and its approaches, Information Retrieval, Semantic Search. Summarization: Summarization, Extractive vs. Abstractive Summarization, Information Fusion, Single and Multi-document Summarization, Question Answering.	
UNIT:5	
Introduction to Chatbot Applications: Introduction to Chatbot Applications, Retrieval-based vs. Conversation-based Chatbots, NLU (Natural Language Understanding) and NLG (Natural Language Generation). Machine Translation: Machine Translation, Interlingua.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1) Daniel Jurafsky and Prentice Hall James H Martin, “Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, Prentice Hall, 2 nd Edition, 2018. 2) C.Manning and H.Schutze, “Foundations of Statistical Natural Language Processing”, MIT Press. Cambridge, MA:, 1999. 3) James Allen, Benjamin Cummings, “Natural Language Understanding”, 2 nd edition, 1995.	
Reference Books: 1) Yoav Goldberg, <i>Neural Network Methods for Natural Language Processing</i> . 2) http://mccormickml.com/2106/04/19/word2vec-tutorial-the-skip-gram-model/ 3) https://nlp.stanford.edu/pubs/glove.pdf	

SUBJECT CODE		SUBJECT NAME						L	T	P	C			
		Formal Language & Automata Theory						3	0	0	3			
Course Educational Objectives:														
CEO 1	To familiarize students to construct regular expressions, regular grammars & to identify non - regular languages													
CEO 2	Teach students to identify context - free languages, to convert the given grammar to various normal forms, & to make use of membership algorithm.													
CEO 3	Teach students to construct Push - Down Automata which represent context - free languages, closure properties, & to identify non - context - free languages.													
CEO 4	To familiarize students to Recursively Enumerable languages, Recursive languages, construction of Turing Machines, PCP, & undecidable problems.													
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>														
CO1	Explain the basic concepts of formal computation and its relationship to language.													
CO2	Classify language into their types and derive its equivalent regular expression.													
CO3	Apply formal reasoning to construct different language and grammar set.													
CO4	Analyze complexity of various problem solving techniques using PDA and TM with recursion.													
CO 5	Analyze the properties of formal languages, grammars, and automata to determine their computational limits and capabilities.													
CO 6	Design and construct deterministic and non-deterministic finite automata, context-free grammars, and pushdown automata to solve specific language recognition problems													
CO-PO & PSO Mapping (Articulation Matrix):														
Cos	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1												
CO2	3	2												
CO3	2	3	2											
CO4	2	3												
CO5	2	3												
CO6	2	3												
Avg.	2.3	2.5	1											
SYLLABUS														

UNIT:1 (12 Hours) Mathematical preliminaries: Alphabet, String, Languages, Grammars, Strings and operations on strings. Finite Automata: Definition, Basic model, Types of Finite Automata, Design of DFA, Design of NFA, NFA vs. DFA, Eliminating ϵ-transitions from NFA, NFA to DFA conversion, NFA or DFA as a language acceptor, Minimization of Finite Automata. Equivalence of two Finite Automata
UNIT:2 (16 Hours) Regular Expressions: Regular Set and Regular Expressions. Operators in Regular expressions, Building Finite Automata from Regular expression, Arden's theorem & Building Regular expression from Finite Automata, Pumping Lemma for Regular languages, Closure properties of Regular languages.
UNIT:3 Grammar: Definition, Regular Grammar, Regular Grammar to Finite Automaton, Finite Automaton to Regular Grammar, Designing Context Free Grammar, String Derivation, Parse Tree Construction, Ambiguous Grammar, Chomsky and Greibach Normal Forms, CYK parsing algorithm, Closure Properties of CFL, Pumping Lemma for CFL, Introducing Non-Context Free Grammar, Chomsky Hierarchy.
UNIT:4 (12 Hours) Push Down Automata: Basic Model, Components, Moves of a PDA, ID of a PDA, Design of Deterministic PDA and Non-deterministic PDA, PDA to CFG and CGA to PDA conversion. Turing Machines: Basic Model, Components, move of a TM, ID of TM, Design of a TM, Variants Of Turing Machine, Recursively Enumerable Languages, Undecidable problems, Post correspondence problem as an Undecidable Problem. Linear Bounded Automata and Context Sensitive Languages
UNIT:5 (08 Hours) Primitive Recursive functions: μ- Recursive functions, Cantor and Godel numbering, Ackermann's function, Excursiveness of Ackermann and Turing computable functions. Church Turing hypothesis Recursive and Recursively Enumerable sets, NP Completeness: P and NP, NP complete and NP Hard problems. Teaching Methods: Chalk& Board/ PPT/Video Lectures Text Books: 1. <i>Introduction to the theory of computation</i>, Michael Sipser, Cengage Learning. 2. <i>Formal Language and Automata Theory</i>, C. K. Nagpal, Oxford University Press. Reference Book: 1. Theory of Computer Science: Automata, Languages and Computation by K L PMishra and N Chandrasekaran 2. Introduction to Automata Theory, Languages and Computation: J. E. Hopcroft, J.D Ullman, Pearson Education.

Subject Code		Subject Code						L	T	P	C				
		Software Project Management						3	0	0	3				
Course Educational Objectives															
CEO1	To outline the need for Software Project Management														
CEO2	To highlight different techniques for software cost estimation and activity planning.														
CEO3	At the end of the course the students will be able to practice Project Management principles while developing software.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Demonstrate project planning, scheduling, tracking, risk analysis and Project cost estimation.														
CO2	Apply the report writing techniques and standards to project report.														
CO3	Administer the progress of projects and to assess & evaluate the risk of project.														
CO4	Identify the factors that influence people’s behavior in a project environment and appropriately select people for the project.														
CO 5	Develop and manage detailed project schedules, including activity sequencing, resource allocation, and risk management, using project management techniques such as Critical Path Method (CPM), PERT, and Monte Carlo simulation to ensure timely project delivery.														
CO 6	Implement effective project control mechanisms, including cost monitoring, earned value analysis, and change control, to track project progress, manage software configurations, and oversee contract management, ensuring successful project completion and stakeholder satisfaction.														
CO-PO & PSO Mappingbp															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2	2												
CO2	3	2	1												
CO3	3	2	2												
CO4	3	2	1												
Avg	3	2	1.5												
SYLLABUS															
UNIT:1 (10 Hours)															
Project evaluation and project planning: Importance of Software Project Management ,Activities Methodologies, types of Software Projects ,Setting objectives ,Management Principles ,Management Control , Project portfolio Management, Cost-benefit evaluation technology , Strategic program Management , Stepwise Project Planning.															
UNIT:2 (12 Hours)															
Project life cycle and effort estimation: Engineering and production stages, inception,															

Elaboration, construction, transition phases. Process Models, Choice of Process models, Rapid Application development , Agile methods, Extreme Programming ,SCRUM ,Managing interactive processes ,Basics of Software estimation: Effort and Cost estimation techniques, COSMIC Full function points , COCOMO II A Parametric Productivity Model ,Staffing Pattern.	
UNIT:3	(14 Hours)
Activity planning and risk management and control : Objectives of Activity Planning: Objectives of Activity Planning, Project Schedules, Activities Sequencing and Scheduling. Network Planning Models: Network Planning Models, Forward Pass & Backward Pass Techniques, Critical Path Method (CPM), PERT Technique. Risk Management: Risk Identification, Assessment, Monitoring. Resource Allocation and Scheduling: Resource Allocation, Creation of Critical Patterns, Cost Schedules.	
UNIT:4	(10 Hours)
Framework for Management and Control: Framework for Management and Control, Collection of Data, Project Termination, Visualizing Progress, Cost Monitoring. Earned Value Analysis and Project Tracking: Earned Value Analysis, Project Tracking, Change Control. Software Configuration and Contract Management: Software Configuration Management, Managing Contracts, Contract Management. Advanced Techniques: Monte Carlo Simulation.	
UNIT:5	(10 Hours)
Staffing in software projects : Managing people , Organizational behavior, Best methods of staff selection, Motivation ,The Oldham, Hackman job characteristic model, Ethical and Programmed concerns, Working in teams, Decision making ,Team structures, Virtual teams , Communications genres ,Communication plans. Process Automation: Automation Building blocks, The Project Environment.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Bob Hughes, Mike Cotterell and Rajib Mall: <i>Software Project Management – Fifth Edition</i>, Tata McGraw Hill, New Delhi, 2012. 2. Robert K. Wysocki “<i>Effective Software Project Management</i>” – Wiley Publication, 2011. 3. Walker Royce: “<i>Software Project Management</i>”- Addison-Wesley, 1998. 4. Gopala swamy Ramesh, “<i>Managing Global Software Projects</i>” – McGraw Hill Education (India), Fourteenth Reprint 2013.	
Ref. Books: 1. Royce, “<i>Software Project Management</i>”, Pearson Education, 1999. 2. Jalote, “<i>Software Project Management in Practice</i>”, Pearson Education, 2002.	

VIII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (AIML) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PE	23BCMPE480**	Professional Elective-VI (From Bucket)	3	0	0	3	40	60
2	OE	23BCMOE480**	Open Elective-IV (From Bucket)	3	0	0	3	40	60
3	OE	23BCMOE480**	Open Elective-V (From Bucket)	3	0	0	3	40	60
TOTAL				9	0	0	9	120	180

PRACTICAL/SESSIONAL									
4	PST	23BCMPE48101	Technical Seminar	0	0	0	1	0	50
5	PST	23BCMPW48101	Major Project Work-II	0	0	6	3	50	50
			TOTAL	0	0	6	4	50	100
			TOTAL	9	0	6	13	170	280

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Sem	Professional Elective VI
1	23BCMPE48011	Big Data Analytics
2	23BCMPE48012	Big Data and Visualization

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective IV & V
1	23BCMOE48011	E-Commerce & ERP
2	23BCMOE48012	Block chain Technology
3	23BCMOE48013	Real Time Systems
4	23BCMOE48014	Embedded Systems
5	23BCMOE48015	Software Testing

Subject Code		Subject Code								L	T	P	C			
		Big Data Analytics								3	0	0	3			
Course Educational Objectives																
CEO1	To outline the need for Big Data															
CEO2	To highlight different techniques for big data analysis.															
CEO3	At the end of the course the students will be able to practice Job Execution in Hadoop Environment principles while developing software.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	Identify Big Data and its Business Implications.															
CO2	List the components of Hadoop and Hadoop Eco-System															
CO3	Access and Process Data on Distributed File System															
CO4	Develop Big Data Solutions using Hadoop Eco System															
CO 5	Analyze Info sphere Big Insights Big Data Recommendations.															
CO 6	Apply Machine Learning Techniques using R.															
CO-PO & PSO Mappingbp																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	2	2													
CO2	2	2	1													
CO3	3	2	2													
CO4	3	2	1													
CO5	3	2	1													
CO6	3	2	1													
Avg	2.8	2	1.3													
SYLLABUS																
UNIT:1 (10 Hours)																
Types of Digital Data, Introduction to Big Data, Big Data Analytics, History of Hadoop, Apache Hadoop, Analysing Data with Unix tools, Analysing Data with Hadoop, Hadoop Streaming, Hadoop Echo System, IBM Big Data Strategy, Introduction to Info sphere Big Insights and Big Sheets.																
UNIT:2 (12 Hours)																
The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures.																
UNIT:3 (14 Hours)																
Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features.																
UNIT:4 (10 Hours)																
Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive : Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions. Hbase : HBasics, Concepts, Clients, Example, Hbase Versus RDBMS. Big SQL : Introduction																
UNIT:5 (10 Hours)																
Machine Learning : Introduction, Supervised Learning, Unsupervised Learning, Collaborative																

Filtering. Big Data Analytics with BigR
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. Tom White “ <i>Hadoop: The Definitive Guide</i>” Third Edit on, O’reily Media, 2012. • 2. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.
Ref. Books: 1. Michael Berthold, David J. Hand, "Intelligent Data Analysis”, Springer, 2007. 2. Jay Liebowitz, “Big Data and Business Analytics” Auerbach Publications, CRC press (2013).

Subject Code		Subject Code										L	T	P	C	
		Ecommerce and ERP										3	0	0	3	
Course Educational Objectives																
CEO1	To outline the need for Ecommerce															
CEO2	To highlight different techniques for ecommerce															
CEO3	At the end of the course the students will be able to manage Ecommerce web sites and ERP.															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Understand fundamental concepts and technologies related to ERP															
CO2	Explain the different phases of ERP implementation life cycle.															
CO3	Examine the modules, benefits and various tools of ERP.															
CO4	Analyze the impact of e-commerce on business model and strategies.															
CO 5	Assess the electronic payment systems and software.															
CO 6	Identify and solve the security issues related to communication.															
CO-PO & PSO Mappingbp																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	2	2													
CO2	2	2	1													
CO3	3	2	2													
CO4	3	2	1													
CO5	3	2	1													
CO6	3	2	1													
Avg	2.8	2	1.3													
SYLLABUS																
UNIT:1 (10 Hours) Overview of enterprise, Related Technologies, Business intelligence, Business process reengineering, Data warehousing, Data mining, Online analytical processing, Product life cycle management, Supply chain management, Customer relationship management, Management information system, Decision support system, Executive information system.																
UNIT:2 (12 Hours) Implementation Challenges, Strategies, Life Cycle, Pre-implementation Tasks, Requirements Definition, Methodologies, Package selection, Project Teams, Process Definitions, Vendors and Consultants, Data Migration, Project management, Post Implementation Activities.																
UNIT:3 (14 Hours) Business models, Revenue models and business processes, Opportunities and nature of ecommerce, Internet protocols, Internet2 and semantic web, Selling on the Web and Marketing on the Web, Business to Business Strategy from Electronic Data Interchange to E-Commerce..																
UNIT:4 (10 Hours) Web hosting alternatives, Basic and advance functions of e-commerce software, Software for small, Mid-size and large businesses, Online payment basic, Payment cards, Electronic cash, Electronic wallets, Stored value cards, Internet technologies and banking.																
UNIT:5 (10 Hours) Overview of online Security issues, Security for client computers, Communication channel																

security, Security for server computers.
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. Alexis Leon, <i>“Enterprise Resource Planning”</i>, 2nd edition, Tata McGraw Hill. 2. Schneider, <i>“E-commerce”</i>, 7th edition, Course Technology of Thomson Learning.
Ref. Books: 1. Ravi Shankar & S. Jaiswal, <i>“Enterprise Resource Planning”</i>, Galgotia 2. Ravi Kalakota, <i>“Frontiers of e-commerce”</i>, 1st edition, Pearson

Subject Code	Subject Code	L	T	P	C										
	Block Chain Technology	3	0	0	3										
Course Educational Objectives															
CEO1	To outline the need for Block Chain Technology														
CEO2	To highlight different techniques for block chain														
CEO3	At the end of the course the students will be able to understand the application of block chain.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Describe the basic concepts and technology used for blockchain														
CO2	Describe the primitives of the distributed computing and cryptography related to blockchain.														
CO3	Illustrate the concepts of Bitcoin and their usage.														
CO4	Implement Ethereum block chain contract.														
CO 5	Apply security features in blockchain technologies.														
CO 6	Use smart contract in real world applications.														
CO-PO & PSO Mappingbp															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2	2												
CO2	2	2	1												
CO3	3	2	2												
CO4	3	2	1												
CO5	3	2	1												
CO6	3	2	1												
Avg	2.8	2	1.3												
SYLLABUS															
UNIT:1 (10 Hours) Need for Distributed Record Keeping, Modeling faults and adversaries, Byzantine Generals problem, Consensus algorithms and their scalability problems, Nakamoto’s concept with Blockchain based cryptocurrency, Technologies Borrowed in Blockchain – hash pointers, consensus, byzantine fault-tolerant distributed computing, digital cash etc..															
UNIT:2 (12 Hours) Atomic Broadcast, Consensus, Byzantine Models of fault tolerance, Hash functions, Puzzle friendly Hash, Collision resistant hash, digital signatures, public key crypto, verifiable random functions, Zero-knowledge systems															
UNIT:3 (14 Hours) Bitcoin blockchain, Challenges and solutions, proof of work, Proof of stake, alternatives to Bitcoin consensus, Bitcoin scripting language and their use.															
UNIT:4 (10 Hours) Ethereum and Smart Contracts, The Turing Completeness of Smart Contract Languages and verification challenges, Using smart contracts to enforce legal contracts, comparing Bitcoin scripting vs. Ethereum Smart Contracts, Writing smart contracts using Solidity & JavaScript.															

AIML

consensus algorithms to prevent these attacks
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. Narayanan, Bonneau, Felten, Miller and Goldfeder, "Bitcoin and Cryptocurrency Technologies – A Comprehensive Introduction", Princeton University Press. 2. Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform, 2017.
Ref. Books: 1. Imran Bashir, "Mastering Blockchain: Distributed ledger technology, decentralization, and smart contracts explained", Packt Publishing. 2. Merunas Grincalaitis, "Mastering Ethereum: Implement Advanced Blockchain Applications Using Ethereum-supported Tools, Services, and Protocols", Packt Publishing.




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